
BBE SCHOOL

Economics Full Course

A learning system for BBE School

BBE Path

Scene

Idea

Mechanism

Practice

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Original teaching for the Economics Full Course — built for exam recognition, not for reading like a traditional textbook reprint.

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Chapter 2

Basic economic concepts

Every purchase, every job offer, and every business idea sits inside the same system of people exchanging scarce resources, and this chapter is a map of that system. You will move from everyday exchange through scarcity and opportunity cost into how economies organise decisions, how markets clear, and why competition matters, until the exam language starts to feel familiar rather than foreign.

Learning objectives

1. Explain how households and businesses participate in exchange, and define what counts as a business.
2. Apply scarcity and opportunity cost to household, firm, and government choices without confusing scarcity with poverty.
3. Distinguish microeconomics from macroeconomics as two lenses on the same scarcity problem.
4. Trace the circular flow among households, businesses, and government, and explain how specialisation and division of labour raise productivity.
5. Compare market, planned, and mixed systems by who decides what, how, and for whom.
6. Use supply, demand, *ceteris paribus*, equilibrium, surplus and shortage, and competition to reason about market outcomes.

2.1 You are already in the economy

Saturday at the campus gate, Mira sells repaired second-hand bikes from a fold-out stand near the tram stop, taking cash or card for a tuned bike while paying a workshop for spare parts and renting storage for the winter. Across the street a household buys tram tickets, groceries, and streaming, and although nobody here is studying economics in the moment, every exchange is the economy at work.

A business is an entity that offers goods and/or services to customers. It does not produce mainly for the owner's private household alone; it serves customers, and it usually charges a price (or another payment) so something of value comes back.

Exchange is the handing over of goods, services, money, or other means of payment so that needs and wants can be met. Money makes exchange easier, but barter - bike-repair lessons for homemade bread - still counts as exchange.

Households supply labour and buy goods and services, while businesses supply goods and services and buy inputs such as labour, materials, and equipment, and people can hold both roles at once: Mira is a household member when she buys food and an entrepreneur when she sells bikes. Businesses also have needs of their own - a bike shop needs tyres, tools, and skilled repair hours - so the line between "who buys" and "who sells" is less a wall than a revolving door.

Needs are the basics people require for living and functioning, such as food, shelter, transport, and care, whereas wants stretch beyond that into preferred extras like a designer helmet or a café brunch. Economics studies both, because scarce budgets force people to rank across both categories rather than treat wants as somehow outside the subject.

Whether something counts as a business depends on offering goods or services to customers for payment, not on how informal the stand looks. A student who occasionally fixes friends' bikes for free is being helpful without operating as a business; the same student who advertises on campus, charges 25 euros per tune-up, and buys parts to resell as part of the service clearly matches the definition used in this course; and two neighbours who swap tomatoes for jam are exchanging without either party running a business unless they systematically offer produce to customers.

It helps to notice one good and one service you bought this week and, for each, who stood on the household side and who stood on the business side of the exchange, because that habit makes the definitions concrete. A common mistake is to assume that if you do not run a company you are somehow outside the economy, yet buying, working, saving, and even choosing not to spend are economic decisions, and there is no real opting out. Statements that restrict exchange to money alone, or that claim only entrepreneurs participate, are usually false; absolute words such as never, only, and always deserve suspicion, because barter and ordinary household purchases still count.

Behind these everyday scenes sits a pressure that never quite goes away: resources are not available in unlimited abundance, so households, businesses, and governments all have to economise - managing limited means among competing uses - which is the bridge into scarcity and opportunity cost.

2.2 Scarcity and opportunity cost

One Saturday, Leila can work a café shift that pays 48 euros for the afternoon, or photograph a local sports final that pays 40 euros but builds her portfolio, and she cannot do both. The clock is the scarce resource, and the choice is the economics.

Scarcity means resources are limited while needs and wants pull in many directions. Time, money, machines, materials, and attention all run out. Scarcity forces allocation choices for households, businesses, and governments alike.

Opportunity cost is the benefit of the next-best alternative forgone when you choose something else. It is not the sum of every rejected option - only the single best option you actually gave up.

Scarcity creates cost even when no receipt is printed, because when you pick option A the benefit of option B disappears, and that lost benefit is the opportunity cost. A founder who "pays herself zero" still forgoes the salary she could have earned elsewhere, and a city that resurfaces a road forgoes the rail link it could have funded with the same budget, so the cost lives in the path not taken rather than only in the cash that changed hands.

Opportunity cost (decision rule)

Opportunity cost of choosing A = benefit of the next-best alternative B that is given up

A = chosen option; B = best realistic alternative left behind (not every rejected dream).

It is easy to confuse scarcity with poverty, yet they are not the same idea. Scarcity is the universal condition of limited means facing competing uses, and it applies to wealthy households and rich countries too, forcing trade-offs even when income is high. Poverty is a severe lack of means relative to basic needs - a social and material condition rather than the definition of scarcity - so poverty implies scarcity, but scarcity does not imply poverty.

Opportunity cost always tracks the next-best path, not a shopping-list total. If Sara has 700 euros for one purchase and values a used laptop highly for study while ranking a washing machine as the next-best alternative for saving laundrette fees, and she buys the laptop, her opportunity cost is the benefit of the washing machine forgone - not laptop price plus washer price, and not every other item she once considered. The same logic catches the self-employment salary trap: Omar can earn 32,000 euros a year as an employed bike mechanic, yet when he starts a repair stall and draws no formal wage in year one the accounting wage may show zero while the opportunity cost of his time is at least the 32,000-euro employment package he refused, because a zero cash wage is not a zero opportunity cost.

If Leila chooses the photography gig, her opportunity cost is the benefit of the café shift she gave up - the 48-euro pay in that story - not the fun of both options at once and not the sum of every weekend plan she skipped. Three mistakes keep returning: adding up all rejected options, saying "free" activities have no opportunity cost, and equating scarcity with poverty so that high-income actors supposedly do not face scarcity. True statements usually name the forgone next-best benefit, while false ones often claim zero cost because no money moved, treat opportunity cost as the price of the item bought, or inflate it into the sum of all alternatives. Scarcity forces choice, and choice creates opportunity cost; later sections show the same logic inside markets, where suppliers forgo better uses of their time when prices are low, and inside government budgets, where one project crowds out another.

2.3 What economics studies

Two headlines can sit in the same morning paper and still ask for different lenses: a local bakery raises pastry prices after butter costs jump, while national unemployment falls and inflation edges up. It is the same economy viewed at different zoom levels, and economics is the toolkit that explains both kinds of story.

Economics studies how individuals (in households) and businesses make decisions to satisfy needs and wants with limited resources. It builds theories to explain observed behaviour and to make careful predictions.

Both branches study the same scarcity problem from different distances. Microeconomics zooms to units - one household, one firm, one market interaction, such as how demand for e-bikes changes if buyers receive a purchase bonus - while macroeconomics zooms to aggregates such as growth, unemployment, interest rates, and the overall price level and

inflation for a country or similar whole. When you ask whether more cafés will open on a street if rents fall, you are in micro territory of local firms and a local market; when you ask whether the country's average price level rose by 2% this year, you are looking at aggregate inflation and therefore at macro; and when you ask how one robotics start-up allocates its seed funding, you remain in micro because the unit of analysis is one business, even if the euro amounts are large. Scale of money alone does not decide the label - the unit of analysis does.

A government raise in the minimum wage that changes one supermarket's hiring plan can bridge both lenses: the national policy sits in a macro conversation about labour-market aggregates, while the supermarket's hiring response is a micro decision by one firm. Labelling anything involving government as automatically macro is a trap, because government can appear in micro through a tax on one product and in macro through a national fiscal stance, just as calling a single expensive purchase "macro" because the price tag is large confuses size with scale of analysis. On an exam, match the statement to the unit of analysis: words like inflation, unemployment, growth, and national price level lean macro, while one buyer, one seller, or one product market lean micro.

Section 2.2 gave you scarcity and opportunity cost as the shared engine; this section names the science that studies those decisions at two scales. Markets in 2.6 are mostly a micro stage, while later inflation talk links to macro aggregates. In short, economics is the study of decisions under limited resources, micro looks at units and their interaction while macro looks at aggregates, and theories aim to explain and predict rather than merely describe.

2.4 Circular flow, money, and specialisation

On Friday workers receive wages from firms, and on Saturday those same people spend part of the wages on pizzas, tram rides, and phone plans, while firms use the sales revenue to pay suppliers and next week's wages and public authorities collect taxes to fund streets, schools, and security. The loop never really stops; it is a circular flow.

Circular flow Households mainly offer labour and receive wages; businesses offer goods and services and receive payment. Government levies taxes and provides goods, services, transfers, and subsidies. Together these streams form a circular flow of goods, services, and money.

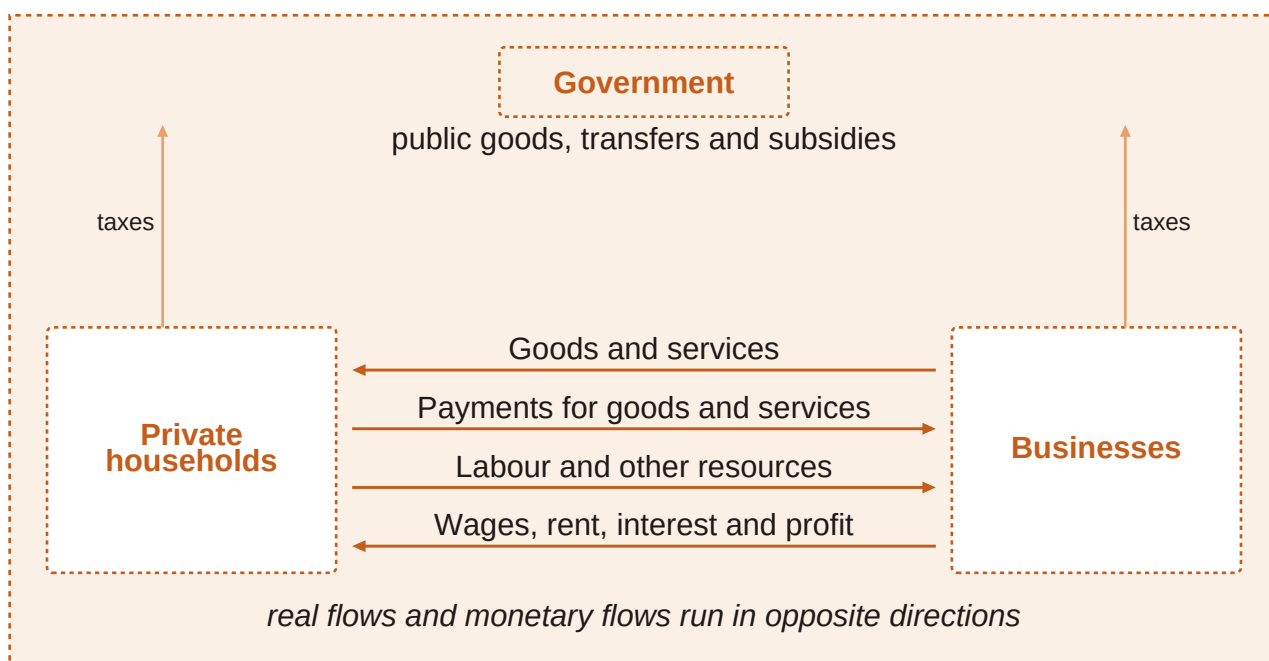


Figure 1. Circular flow: households, businesses, and government exchanging labour, goods/services, taxes, transfers, and money payments.

Money beats pure barter because without it trade needs a double coincidence of wants - each party must want what the other offers at the same time - whereas money works as a medium of exchange that adds flexibility, a unit of account for expressing value, and a store of value for holding purchasing power over time, and those functions work best when money's value stays reasonably stable. When the general price level rises through inflation, the same money buys fewer goods and services so purchasing power falls, and price indexes measure that rise; very high inflation erodes trust in money as a store of value and pushes people to spend it quickly, while low, stable inflation is more compatible with money doing its job.

Government sits in the loop as well: public authorities tax households and businesses, then provide infrastructure, defence, police, and often large parts of health care and education. Some goods are hard for private firms to sell profitably because free riders cannot be excluded from enjoying them, so government provision financed by taxation fills the gap that pure market sale leaves open.

Division of labour and specialisation Exchange lets people and firms concentrate on what they do best instead of producing everything themselves. Specialisation appears inside households, inside firms (departments), between firms on the same or different production stages, across sectors, and across countries with different resources and know-how.

Specialisation shows up at every scale: inside a household one person may shop while another cooks so each can focus; inside a firm procurement, production, sales, accounting, and HR become distinct tasks; between firms wood becomes boards, then furniture, then a retail product, or firms settle into narrow niches; and internationally different climates, resources, labour costs, and legal frameworks shape who produces what. Without trade a household must grow food, sew clothes, repair bikes, and teach itself every skill, which is slow and inefficient, whereas with money and markets the household can sell labour hours and buy specialised goods and services. A bike shop specialises in repairs and a bakery specialises in bread, and both gain from focusing, so specialisation raises output and variety

even as it ties people more tightly to exchange networks.

If a country suddenly could not import any electronics, local retailers would lose specialisation advantages that rested on global supply chains, and new opportunity costs would appear as scarce domestic capacity was pulled toward making what imports once covered. Specialisation is not only upside: highly specialised work can become boring, skills may be hard to redeploy, and a firm brilliant in one niche is exposed if that niche collapses. Circular-flow questions often test who pays whom among wages, taxes, and purchases; money-function questions test medium of exchange, unit of account, and store of value; and specialisation questions test efficiency gains alongside the flexibility risk. Circular flow is the stage on which later sections play: economic systems decide how much of that stage is directed by markets versus by planners, and supply and demand explain prices inside the market channels of the flow.

2.5 Economic systems: who decides?

Imagine a tonne of steel and three ways to allocate it. In one system private firms bid and households' spending patterns pull production toward cars or bridges; in another a planning office assigns the tonne to a factory list; and in a third markets do most of the steering while government taxes, regulates, and funds social and environmental goals. Same scarce steel, different decision rights.

An economic system answers the allocation questions: what is produced, how it is produced, and for whom - by assigning decision power to markets, to planners, or to a mix of both.

In a market economy individuals and businesses make many of their own economic decisions, with prices and private ownership playing a large role, whereas in a planned system government mainly or partly decides which goods and services are offered - and often at which prices - while controlling resources and means of production, so job and product choice for people is more limited. Most real countries sit somewhere in between as mixed systems in which markets allocate widely while government sets legal frameworks and may support the poor, protect the environment, or provide major public services, sometimes called social or eco-social market economies when the mix is deliberate.

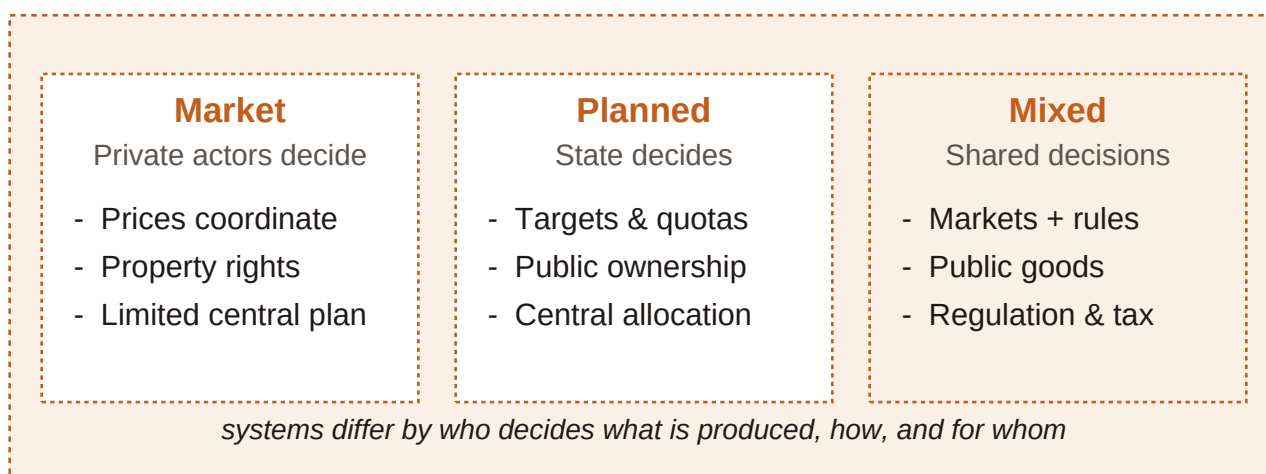


Figure 2. Spectrum of economic systems: planned coordination at one end, freer market coordination at the other, with mixed systems in between.

Question	Market-leaning	Planned-leaning	Mixed (typical modern case)
What is produced?	Largely guided by demand, prices, and private profit/opportunity	Mainly set by government plans and targets	Markets dominate many goods; public priorities steer others
How is it produced?	Firms choose methods and organise resources	Authorities control resources and production means (mainly/partly)	Private methods plus regulation, standards, and public providers
For whom?	Income and market exchange shape access	Allocation and rationing more centrally directed; limited consumer/job choice	Markets distribute much; taxes/transfers and public services reshape access
Government role	Can be thin (legal framework) in freer variants	Dominant in production and resource control	Active but not total - legal rules, social goals, environment, public goods

Table 1. Comparing economic systems by decision rights

You can label the system by the signal rather than the slogan. Private cafés that freely set menus and prices while customers walk away if unhappy point to a market mechanism; a ministry that assigns quarterly output quotas and fixed retail prices for staples points to a planned mechanism; and private firms that compete while government funds schools, taxes pollution, and runs a safety net point to a mixed arrangement. Judge by who holds the

decision rights on what, how, and for whom. If a country privatises shops but still sets fuel prices and owns the railways, it is drifting toward the mixed column even while some decisions remain planned, and it helps to remember that "market economy" never means zero government - even freer market systems rely on law - just as every mixed economy is not identical, because the mix can be light or heavy. Transformation stories about former planned systems adopting market principles test whether you notice a shift in decision rights, not a change in geography alone. Systems set the rules of the game, and markets - next - are the coordination device that market-leaning and mixed systems use for many goods and services.

2.6 Supply, demand, and market equilibrium

Parents message tutors for evening maths help, and at low hourly rates many families want hours while few skilled tutors bother to log on, whereas at very high rates tutors flood the platform but fewer families book. Somewhere in the middle the hours people want line up with the hours tutors offer, and that meeting point is the market at work.

A market is where buyers and sellers communicate the terms of exchange. It can be a physical place (a flea market) or virtual (an online platform), and it can specialise (labour, housing, money, capital, commodities, consumer goods).

Supply is the quantity of a good or service available for purchase. For most goods, a higher price raises the quantity supplied, *ceteris paribus* - the law of supply. Capacity, resources, and opportunity cost of the seller's time all matter.

Demand is the quantity customers are willing and able to buy. For most goods, a higher price lowers quantity demanded, *ceteris paribus* - the law of demand. Willingness to pay links to the utility (satisfaction) people get from the good or service.

Ceteris paribus Latin shorthand for "all other things held constant." Supply and demand laws isolate the price-quantity link while freezing other influences. When those other influences change, curves shift.

The supply curve slopes up because higher prices pull more sellers in and encourage more hours from existing sellers, while opportunity cost explains the floor: if tutoring paid less than a tutor's next-best use of time, skilled people leave. Rising marginal cost - the extra cost of one more unit - also pushes firms to need a higher price before expanding output.

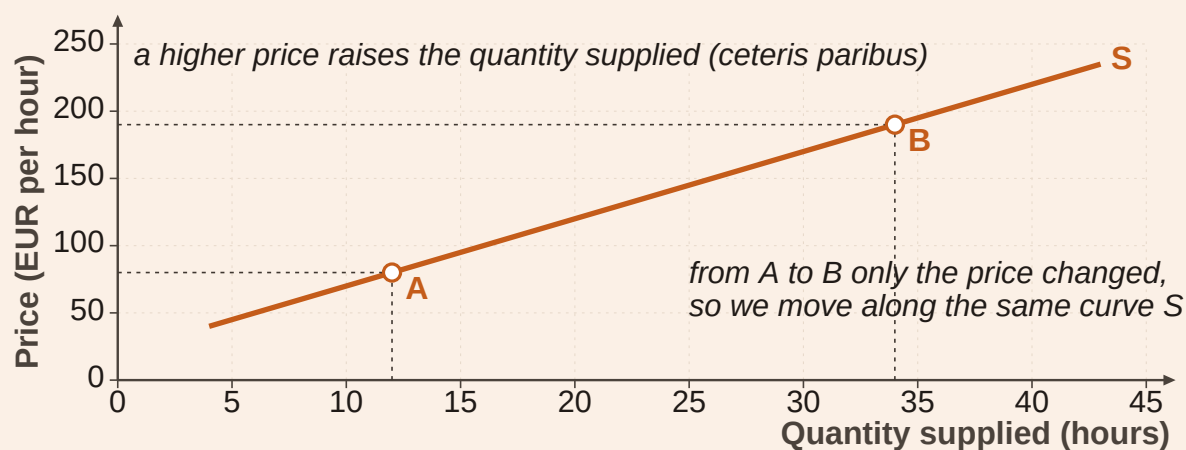
Supply hours of tutoring offered per week

Figure 3. Supply curve for online maths tutoring (illustrative): price (euros per hour) on the vertical axis, quantity (hours per week) on the horizontal axis - upward sloping.

The demand curve slopes down because as price rises more buyers cannot or will not pay, cutting hours or switching to substitutes such as group classes, apps, or self-study, whereas as price falls quantity demanded rises. Price and quantity demanded move in opposite directions when other things are held constant.

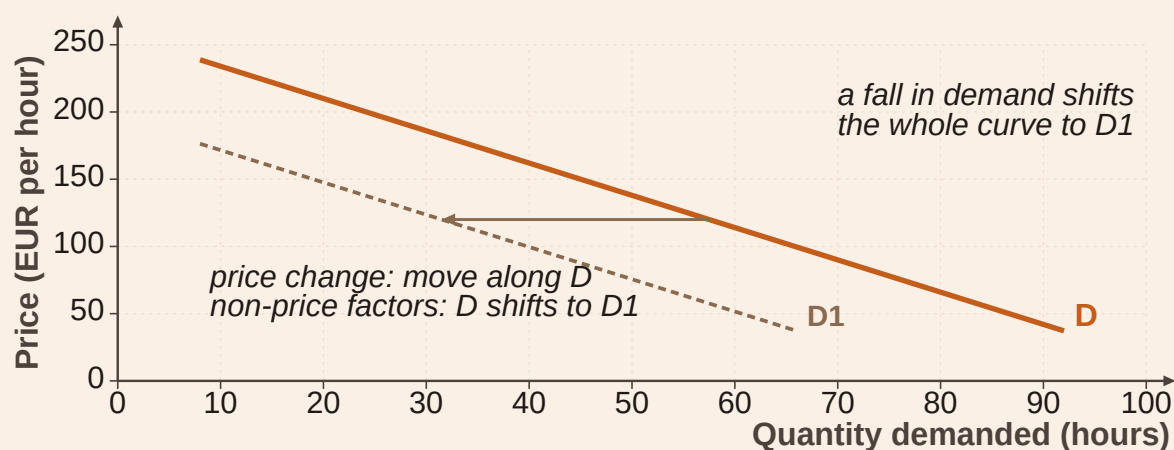
Demand hours of tutoring bought per week

Figure 4. Demand curve for online maths tutoring (illustrative): higher price, lower quantity demanded - downward sloping.

Market equilibrium condition

Equilibrium: quantity demanded = quantity supplied at the equilibrium (market) price

At that price there is neither surplus ($Q_s > Q_d$) nor shortage ($Q_d > Q_s$).

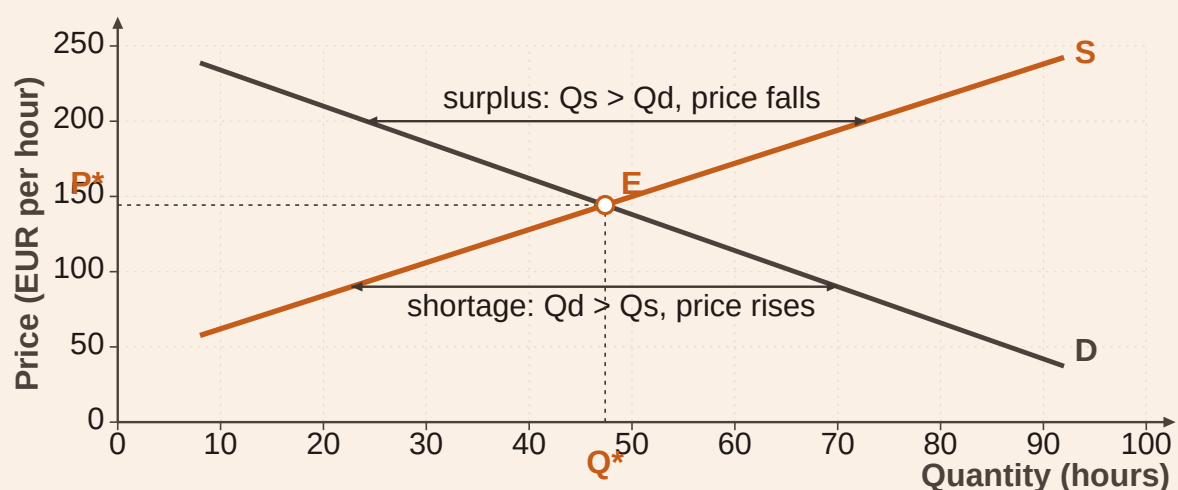
Market equilibrium supply and demand in one market

Figure 5. Supply and demand together: intersection is equilibrium price and quantity. Above it, surplus; below it, shortage.

Price (euros/hour)	Quantity demanded (hours/week)	Quantity supplied (hours/week)	Market signal
18	620	200	Shortage - $Q_d > Q_s$
24	500	320	Shortage
30	400	400	Equilibrium
36	300	480	Surplus - $Q_s > Q_d$
42	220	560	Surplus

Table 2. Illustrative tutoring market schedule (original numbers)

Reading the tutoring schedule, at 30 euros quantity demanded equals quantity supplied at 400 hours, which is the equilibrium price and quantity; at 36 euros tutors offer 480 hours but families only want 300, leaving a surplus of 180 hours and pressure for price to ease downward; and at 18 euros families want 620 hours while only 200 are offered, leaving a shortage of 420 hours and pressure for price to rise. Disequilibrium creates surplus or shortage, and equilibrium is the balance point where both sides meet.

A movement along a curve is caused by a change in the good's own price, sliding you to a new point on the same curve - for example, when the tutoring fee falls from 36 to 30 and you move along demand to a higher quantity. A shift of the curve is caused by a non-price determinant that breaks the ceteris paribus freeze, so the whole curve moves left or right - for example, when household incomes rise and demand shifts right at every price. Demand

shifters include income, preferences, complementary goods, and substitutes; supply shifters include the number of suppliers, technology, resource prices, and price expectations; and other things equal, a rightward demand shift tends to raise equilibrium price and quantity while a rightward supply shift tends to lower price and raise quantity.

When parents' incomes rise and they book more tutoring at each price, the demand curve shifts right; when a new app trains more tutors quickly, the supply curve shifts right; and when only the platform's listed hourly fee changes while preferences stay unchanged, you are looking at movement along the curves rather than a shift story. The diagnostic question is whether the own-price changed or a background condition changed. The same logic helps explain a classic inflation story: if the quantity of money rises and people can spend more, demand for goods can increase, and if goods available do not rise in step, prices climb as too much money chases too few goods, while raising the price of borrowing money through interest rates can cool spending and ease price pressure - a bridge from micro curves to a macro outcome.

Not every change is a shift: when the good's own price changes you move along a curve, and when a background determinant changes the curve itself moves. Mixing surplus with shortage language, or ignoring *ceteris paribus* so that the "laws" look broken because a second variable moved, will also lead you astray.

On exam items, draw or imagine the graph and ask whether price changed (movement along) or a determinant changed (shift). At a stated price, compare quantity demanded with quantity supplied to name surplus or shortage, remembering that equilibrium is the intersection story rather than any price sellers happen to like.

Opportunity cost from 2.2 explains why supply dries up at low prices, circular flow from 2.4 is where these market trades sit, and competition in 2.7 changes how much power sellers have over the price you just analysed. The law of supply says that a higher price raises quantity supplied when other things are held constant; the law of demand says that a higher price lowers quantity demanded under the same freeze; equilibrium equates the two quantities while other prices create surplus or shortage; and own-price causes movement along a curve while other factors shift it.

2.7 Competition in the market

After a steep hike there is a single hut selling soup, and hikers pay because walking another hour for alternatives is painful, whereas down in the town five coffee shops sit on one street so a price rise at one shop sends customers next door. Same product category, very different competitive heat.

Competition The intensity of competition rises when more suppliers sell in the market and when substitute goods are easy to find. More rivals and closer substitutes mean less room for any one seller to dictate terms.

Markets stretch from monopoly toward many sellers. One supplier is a monopoly - rare in pure form, though possible for a national railway or a local-only seller such as the only bar inside a theatre. A few large suppliers form an oligopoly, as with telecoms or car makers, where rivals watch each other closely and illegal cartels that fix terms to kill competition are

generally not allowed. Many buyers and sellers with no single player able to set the price approach the theoretical benchmark of perfect competition, which asks for full information for all players, free entry and exit, and no personal preferences because goods are replaceable across sellers. Real markets seldom match that checklist perfectly, though standardised goods such as some agricultural products can come closer, and even with fewer sellers rivalry can still be fierce if competitors react aggressively.

Competitive pressure weakens when there are few suppliers or only one, when substitutes are weak, and when local demand is captive - as with the summit hut or the theatre bar - because the seller then has more pricing room. It strengthens when many suppliers offer close substitutes and buyers can switch easily, which is why laws often aim to protect rivalry for customers' benefit. Ranking pressure by counting suppliers and checking substitutes, a city tram platform that alone is licensed to run the system looks monopoly-like; four mobile networks watching each other's tariffs look like oligopoly; dozens of nearly identical grain sellers with easy entry sit nearer to perfect competition; and tutoring apps that multiply while video courses substitute for live hours intensify competition through both more suppliers and better substitutes.

If a village has two bakeries and a supermarket opens with fresh bread daily, both the number of suppliers and the availability of substitutes rise, and each bakery's pricing room shrinks as buyers gain easier alternatives. Monopoly is less common than people assume - the course stance is that monopolies are rare in market economies, though local monopoly-like pockets exist - and oligopoly should not be read as "no competition," because rivalry can be intense, while cartel-style agreements among rivals are usually illegal rather than harmless. If an exam stem adds suppliers or substitutes, competition rises; if it isolates a single local seller with no nearby alternative, think monopoly-like power; and perfect-competition answers usually need the theoretical conditions, not merely "many shops." Competition shapes the supply side you drew in 2.6: weak competition can hold price above a more rivalrous outcome, while strong competition disciplines sellers, which is why policy often protects rivalry.

Chapter recap

- Households and businesses exchange to meet needs and wants, a business offers goods or services to customers usually for a price, and nobody fully opts out of economic decisions.
- Scarcity forces allocation, opportunity cost is the next-best alternative forgone, and scarcity is not the same as poverty.
- Economics studies decisions under limited resources, with micro looking at units and interactions and macro looking at aggregates such as growth, unemployment, and inflation.
- Circular flow links households, businesses, and government; money eases exchange; and division of labour enables specialisation with efficiency gains and flexibility risks.
- Market, planned, and mixed systems differ by who decides what is produced, how, and for whom.

- Supply and demand meet at equilibrium under ceteris paribus; own-price causes movements along curves while other factors shift them; surplus and shortage flag disequilibrium; and more suppliers and substitutes intensify competition, while pure monopoly remains rare.

Chapter 3

Focus on different types of businesses

Chapter 2 showed how households and businesses meet in markets; Chapter 3 asks a sharper question about what you do once you can see a business at all - namely, how you describe it accurately. Two firms can both sell to customers and still differ in the resources they combine, the sector they sit in, whether profit is the mission, how large they are, and how far their market reaches. Once you overlay stakeholder interests on that map, you can classify almost any organisation the exam throws at you.

Learning objectives

1. Name the factors of production (land, labour, capital, entrepreneurship/enterprise) and sort concrete inputs into the right factor.
2. Classify activity as primary, secondary or tertiary, and explain why developed economies are tertiary-heavy.
3. Distinguish profit-oriented businesses from not-for-profit organisations without claiming either can ignore cash.
4. Measure business size with employees, turnover and balance-sheet total, and place a firm as micro, small, medium or large using course thresholds.
5. Label a firm as local, national or international/multinational and spot the coordination challenges that grow with reach.
6. Map stakeholders versus shareholders, separate internal from external groups, and recognise conflicting interests.

3.1 Factors of production: what every business combines

At HarborGlow's workshop, weatherproof lanterns for marina shops look like a single finished product, but behind each one sits a mix of resources that rarely announces itself on the shelf: copper sheet arrives from a metals dealer, a cutter shapes the housings, assemblers fit the LEDs, a founder negotiates shop shelf space, and a design spreadsheet decides which models are worth keeping. The lantern may look simple; the resource mix behind it is not.

Factors of production are the resources a business combines to create goods and services. In this course the core set is land, labour, capital and entrepreneurship (also called enterprise). Knowledge and technology reshape how those factors work together.

Those four core factors fit together in a particular way: land supplies natural resources, labour supplies human effort and skill, and capital supplies produced means of production together with the financial resources that keep operations running, while entrepreneurship - or enterprise - is the organising factor that combines the others, chooses what to make, and bears the risk that the plan fails. Without enterprise, land, labour and capital sit idle or misaligned, which is why classification tasks always ask not only what resources are present but who is coordinating them.

In course language, land covers natural resources used in production - soil, water, minerals, forests, fisheries, and sites with natural character - so HarborGlow's coastal workshop site

and the copper ore's origin before refining sit in that category. Labour covers all human resources applied in production, manual and mental, permanent and seasonal, which means cutters, assemblers, a bookkeeper and a quality checker all count. Capital covers machinery, plant, vehicles and the financial resources used to operate, including HarborGlow's press, leased laser cutter, delivery van and cash for payroll. Entrepreneurship or enterprise brings land, labour and capital together, coordinates decisions and bears business risk - the founder choosing the model range, signing shop contracts and absorbing the risk of unsold stock.

Labour is wider than shop-floor muscle: planners, accountants, coaches and call-centre staff are labour whenever they supply human resources. Capital includes equipment whether owned or leased, plus inventories and cash used to run the business. Land remains natural-resource based, so standing forests and mineral deposits count, whereas a finished oak barrel or cut timber ready for milling is an input that has already left the pure land category and is treated with capital or materials logic in classification tasks.

Entrepreneurship and enterprise Entrepreneurship is the activity of organising production and carrying risk. Enterprise is the same organising factor viewed as the venture that combines resources. Exam wording may use either label; both point to coordination and risk-bearing, not to every clever task performed by an employee.

Knowledge and technology sit alongside the four core factors rather than replacing them: fermentation know-how, coding skill or irrigation software changes how land, labour and capital combine, but knowing how to blend wine is knowledge, whereas deciding which vineyard contracts to sign and accepting the loss if the vintage fails is entrepreneurship.

Different businesses emphasise different mixes. On an alpine dairy pasture, meadow and water are land, milkers and cheese makers are labour, vats and a refrigerated truck are capital, and the owner choosing herd size and cheese contracts supplies entrepreneurship. A city coding studio in rented rooms may have little agricultural land, yet location still matters; developers are labour, laptops, licences and a cash buffer are capital, founders pitching clients and carrying unpaid invoices supply entrepreneurship, and stack knowledge with tools sits under knowledge and technology. A printed-circuit plant involves land and natural resources through its site and process water, engineers and operators as labour, lines, clean rooms and working capital as capital, and plant leadership coordinating orders and risk as entrepreneurship. Viable production almost always needs more than one factor at once.

If a technician replaces a phone screen using a spare part from inventory, the technician is labour and the spare part is capital, while entrepreneurship belongs to whoever organises the venture and bears its risk - for example a franchise owner deciding which repairs to advertise this month. A common mistake is to treat labour as only physical work, capital as only owned machines, or entrepreneurship as any purchase decision; labour includes office and service work, leased tools remain capital, and spending money alone is not entrepreneurship, because organising factors and bearing business risk is. Exam vignettes about a winery, repair shop, factory or insurer usually ask whether a named item is land, labour, capital or entrepreneurship, so translate the item into the factor definition first and watch for false splits such as "services use only labour" or "risk-bearing is labour because it takes time." Factors of production are the micro building blocks that later feed sector classification - what the firm mainly does - and stakeholder analysis - who supplies or depends on those factors.

3.2 Primary, secondary and tertiary sectors

Follow one metal through three businesses and the sector labels change even though the material story stays continuous: copper leaves a mine as ore, a cable plant draws it into wire, and a grid-services firm installs and maintains household connections. The same material can therefore sit inside three different sector labels, because sector follows the firm's main activity rather than the material's origin alone.

The three-sector model sorts economic activity into primary (extraction of raw materials), secondary (manufacturing transformation into goods) and tertiary (services).

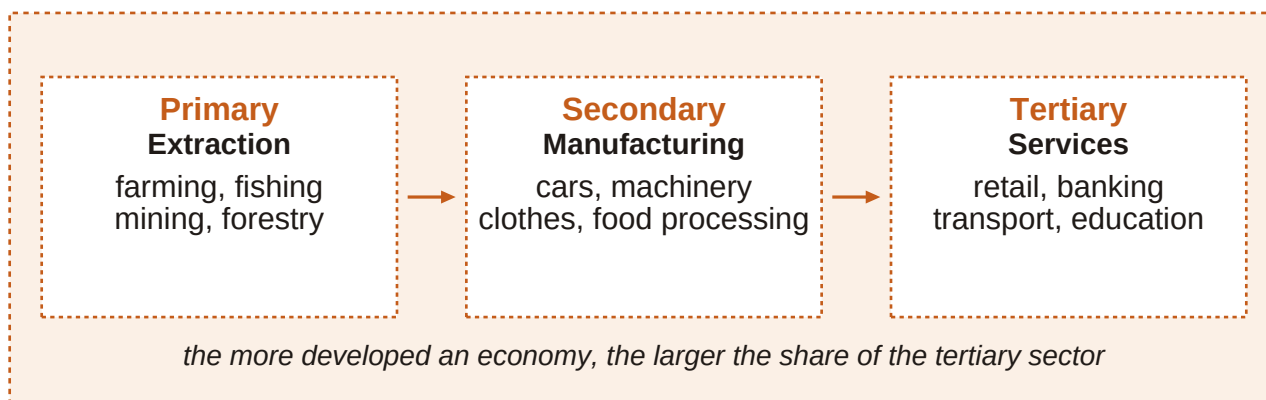


Figure 6. Economic sectors - primary extracts, secondary manufactures, tertiary serves. Follow the main activity of the business, not the story of the raw material alone.

To assign a sector, ask what the business mainly does day to day. If it extracts or harvests from nature, it is primary - farming, fishing, mining, forestry. If it transforms materials into goods, it is secondary - cars, ships, machinery, circuit boards, clothing, computers. If it provides services such as distribution, banking, insurance, coaching, technical support or retail, it is tertiary. A retailer of packaged food is tertiary even though the food began on a farm, because the firm's main operation is distribution and service rather than extraction or fabrication.

Primary and secondary activity usually leave a tangible output at the point of production, and they often take a larger share in less developed economies, especially primary extraction. Tertiary activity delivers services rather than extracting or fabricating, yet it can still use heavy capital and skilled labour; it includes trade, finance, coaching, support and logistics services, and it dominates output in highly developed economies.

As economic development advances, the tertiary sector usually grows in importance. In highly developed economies - for example EU countries with high GDP per capita - the tertiary sector commonly accounts for more than 70% of economic output, while the primary sector is a small percentage of output even though food production remains vital.

Gross domestic product (GDP) GDP is the total monetary value of final goods and services produced within a country's borders in a period (usually a year). It summarises overall economic activity and, when adjusted for inflation and rising over time, signals economic growth.

GDP is useful for comparing activity and tracking growth, but it is imperfect for well-being: not all income sources are captured, quality and sustainability of growth are not scored, and

rebuild spending after an ecological disaster can raise GDP even when people are worse off. Still, GDP often correlates with indicators such as health status and happiness, which is why the measure remains central even with those limits in view.

Walk the copper chain again with that rule in mind: a copper mine extracting ore is primary, a plant turning copper into cables is secondary, and a utility installing and maintaining household grid links is tertiary, while a supermarket selling packaged food is tertiary as distribution and service rather than primary farming. Classify the firm's main operation, and do not drag every downstream seller back into the primary sector because inputs once grew on land. A software firm that writes warehouse apps for manufacturers is tertiary because it sells a service; it does not become secondary merely because its customers are manufacturers. Labelling retail or logistics as secondary "because goods are involved," or assuming primary activity disappears in rich countries, misses the same point: goods can move through tertiary channels, and primary can be small in GDP share while remaining socially essential. Exam chains of mine -> factory -> installer -> retailer usually want one true sector label per firm, and developed-economy items often test the pattern that tertiary exceeds 70% of output together with the idea that GDP growth is not the same as well-being.

3.3 Profit-oriented versus not-for-profit

PulseFit runs paid small-group training in a converted warehouse and expands only when expected surplus justifies new coaches, while RiverAid collects donations and membership fees to fund flood-response kits and puts any surplus into more kits and training rather than private owner dividends. Both organisations need cash; only one treats owner profit as the central success metric.

A profit-oriented business aims for revenues above costs and expenses. Profit rewards owners/investors for risk and can be reinvested to strengthen durability and competitiveness.

Not-for-profit organisation (NPO) A not-for-profit organisation mainly aims to cover costs while pursuing a mission. It still needs revenues - often donations, fees, grants or membership income - and may generate a surplus, but that surplus is reinvested into services or projects rather than treated as private profit as the organisation's purpose.

Surplus still matters for NPOs because covering costs is not optional: without enough inflow, an NPO cannot deliver aid, culture or environmental work, and a surplus is useful precisely because it funds more projects and stronger delivery. The difference from a profit-oriented firm is the primary aim and how success is judged - mission delivery plus financial sustainability versus financial return alongside viability.

Aspect	Profit-oriented business	Not-for-profit organisation (NPO)
Primary aim	Earn profits (revenues above costs) as a central goal	Cover costs while pursuing a mission
Use of surplus	Reward owners/investors for risk; reinvest to strengthen the firm	Reinvest to enhance services or projects; not distributed as private profit goal
Need for revenue	Sales and other commercial income required	Donations, fees, grants or membership income still required
Success focus	Financial return alongside viability and competitiveness	Mission delivery and financial sustainability
Typical examples	Private firms, commercial partnerships, profit-seeking corporations	Charities, humanitarian NGOs, many cultural and environmental associations

Table 3. Profit-oriented businesses and not-for-profit organisations compared

A private language school that expands campuses when forecast profit covers rent risk is profit-oriented, whereas an international relief organisation that raises donations for emergency shelters, covers overheads and uses excess to expand programmes is not-for-profit. Both monitor cash weekly; only the school treats owner profit as a core success score, so cash discipline is shared while purpose and surplus destination separate the two types. If an environmental NPO sells branded water bottles to fund river clean-ups and ends the year with a surplus, that surplus does not automatically turn it into a profit-oriented business, provided the surplus continues to strengthen mission delivery rather than private owner return. Believing NPOs need no money, or that any surplus proves the organisation is secretly profit-oriented, confuses means with purpose; conversely, a firm that reinvests heavily can still be profit-oriented if owner return remains a central aim. Look for aim language - "reward investors" versus "cover costs / mission" - and remember that needing revenue is not the same as being profit-oriented. Profit orientation also shapes which stakeholders push hardest, investors seeking return versus donors seeking mission proof, which Section 3.6 develops.

3.4 Business size: SMEs and large firms

Three bike businesses can share one street name in conversation - a two-person repair loft, a regional frame workshop with dozens of staff, and a listed group with thousands of employees worldwide - and customers may say "bike company" for all three, yet size measures force a clearer label once headcount and financial ceilings enter the picture.

SME (and MSME) SME (small and medium-sized enterprise), sometimes written MSME to highlight micro firms, is the umbrella for businesses below large-enterprise ceilings. Size

matters for access to finance and support programmes, and often for how heavy accounting rules become.

This course uses European Commission-style thresholds that combine staff headcount with either turnover or balance-sheet total. About 99% of EU businesses are SMEs, and crossing the ceilings moves a firm into the large category. In many countries, smaller firms may use simpler accounting methods to calculate profit or loss, a theme Chapter 6 returns to.

Company category	Staff headcount	Turnover	OR Balance sheet total
Micro	< 10	<= EUR 2 m	<= EUR 2 m
Small	< 50	<= EUR 10 m	<= EUR 10 m
Medium-sized	< 250	<= EUR 50 m	<= EUR 43 m
Large	Above SME ceilings	Above SME ceilings	Above SME ceilings

Table 4. Course size categories (EU SME-style thresholds)

Size classification checklist

Category = f(staff headcount, AND (turnover OR balance-sheet total))

Compare headcount to the staff ceiling, then check whether turnover or balance-sheet total stays within the matching financial ceiling. Medium-sized uses <= EUR 50 m turnover or <= EUR 43 m balance-sheet total with staff < 250. Large means above the SME ceilings.

Studio A with 6 employees and EUR 1.2 m turnover sits as micro; Studio B with 40 employees and EUR 8 m turnover is small; Factory C with 180 employees, EUR 40 m turnover and EUR 30 m balance-sheet total is medium-sized; and Group D with 8,000 employees worldwide is large. A-C sit inside SME bands, subject to full legal tests in real programmes, while D is large, and SME innovation support typically targets the SME side of that line. A firm with 45 staff and EUR 12 m turnover fails the small turnover ceiling of <= EUR 10 m, so you test the medium-sized band next rather than calling it small by headcount alone. Using only employees, or treating "SME" as a vibe that the firm "feels small," misses that headcount and financial ceilings both matter, and that medium-sized balance-sheet total peaks at EUR 43 m in the course table rather than EUR 50 m. Numeric vignettes usually ask you to place the firm before debating policy, so memorise the micro, small and medium ceilings and the idea that size affects funding access and accounting burden.

3.5 Local, national and international reach

A corner bakery that sells to walkers within a few streets, a domestic insurer that writes policies in every major city of one country but nowhere abroad, and a components maker that runs plants on two continents and ships to industrial buyers worldwide are all businesses, yet geography is not decoration: it changes logistics, law, language and funding pressure.

Local (regional) business A local or regional business operates in a limited area. Most customers live nearby. It does not serve a national or international market. Typical early-stage challenges include winning enough local customers and avoiding undercapitalisation - own funds are thin and extra finance is hard to obtain.

A national business serves the home-country market but not foreign markets. Location still matters, and the supply chain lengthens because goods and services must reach other places within the country.

International / multinational business An international or multinational business makes and/or sells goods and services in more than one country. Firms often expand abroad when the home market is too small. Cross-border reach brings longer supply chains, different legal and economic systems, cultures, languages and sometimes currencies.

Wider reach raises coordination load because each step outward adds interfaces: local firms fight for nearby demand and cash, national firms must design domestic logistics and consistent delivery, and international firms add borders with rules, cultures, languages and currency risk. As more firms operate across borders, markets and production integrate more deeply - the process called globalisation.

Globalisation refers to the deepening cross-border integration of markets, production and information as more firms operate internationally.

Local reach means a small limited area and nearby customers, with funding pressure and undercapitalisation as classic frictions; national reach means the whole home country and no foreign market, with longer domestic supply chains and multi-location delivery as the typical challenge; international or multinational reach means making and/or selling in more than one country, with longer cross-border supply chains, multiple legal and economic systems and cultures, and possible multi-currency operations that help fuel globalisation.

A neighbourhood bike workshop with clients within a few kilometres is local; a supermarket chain in every major city of one country and none abroad is national; and a components manufacturer with plants in Europe and Asia selling worldwide is international or multinational. The same product family can sit at any reach level, because the market map decides the label. An online tutor who teaches only students in one country remains national for Chapter 3 reach classification even if servers and payment processors sit abroad, because the market definition that matters most is where customers are served rather than where every technical component sits. Calling any firm with an imported input "international," or treating multinational as a synonym for "large," confuses inputs and size with reach: reach is about where the firm makes and/or sells, so a micro exporter can be international while a huge domestic chain can remain national. Vignettes usually stress customer geography and plant locations; match local, national or international definitions, then mention the matching challenge - undercapitalisation, domestic logistics, or cross-border complexity. Wider reach also multiplies stakeholders across communities and governments, which the next section takes up.

3.6 Stakeholders, shareholders and conflicting interests

At MeadowMill, which processes dairy for supermarket buyers, owners want faster fulfilment and night shifts would please buyers while raising utilisation, yet residents hate lorry noise after 22:00, tired staff worry about safety, and a river community watches wastewater quality. One decision, several legitimate interests - that is stakeholder reality.

A stakeholder is anyone who is (potentially) affected by a business's activities and/or has an interest in what the business does. Globalisation and stronger environmental expectations have widened the set of people and groups firms must consider.

A shareholder is an owner of shares in a corporation - a specific ownership stake. Shareholders are stakeholders, but not all stakeholders are shareholders. Employees, customers, suppliers, neighbours and regulators can have strong interests without owning shares.

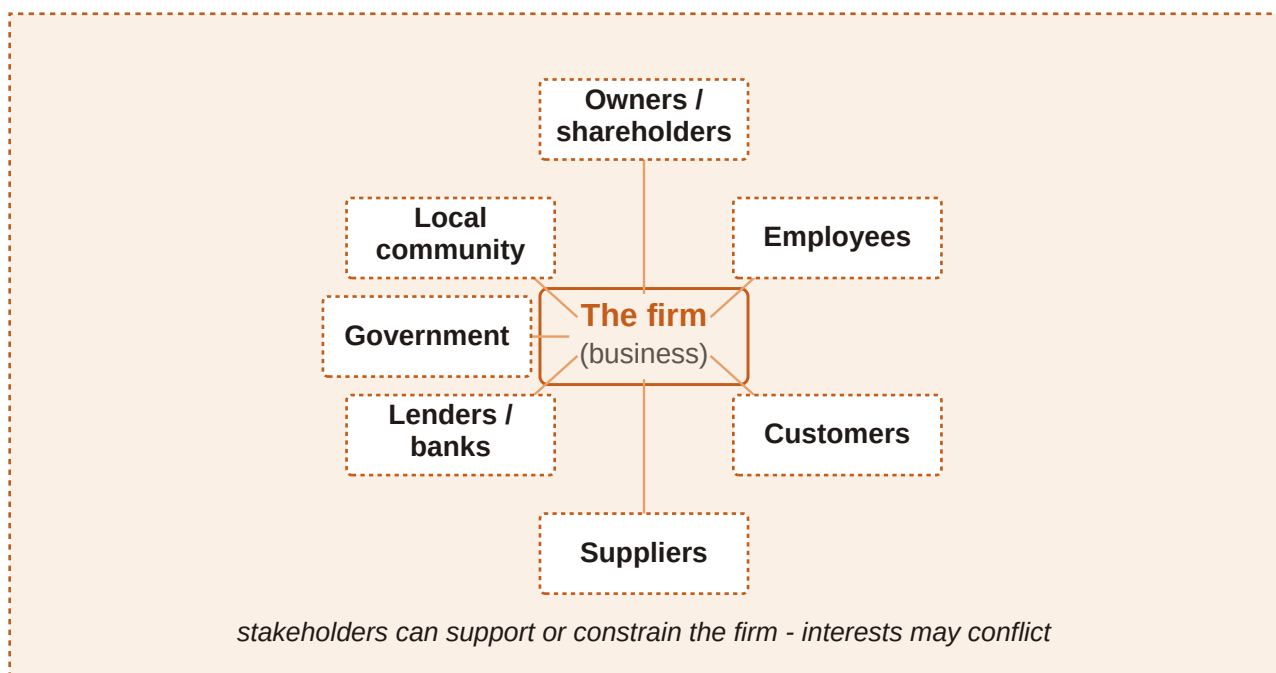


Figure 7. Stakeholder map - the business at the centre, with owners/shareholders, managers, employees, customers, suppliers, government, communities and the natural environment around it.

Internal stakeholders operate inside the organisation - owners and shareholders, managers and employees - while external stakeholders sit outside day-to-day membership but are affected by or interested in the firm: customers, suppliers, government, local, national and international communities, and the natural environment. Managers may or may not also be owners.

Owners, shareholders and investors care about return for risk, value growth and sometimes mission and social contribution; managers care about income, career and the ability to implement strategy; employees care about wages, job security, meaningful work and shared values. Outside the firm, customers want useful, reliable offers at acceptable prices and an ongoing relationship; suppliers want timely orders, fair payment and stable future demand; government wants tax compliance, observance of rules and contribution to public aims; communities care about jobs, congestion, pollution, cultural sponsorship and licence to operate; and the natural environment, as an affected system, points toward sustainable

resource use and lower emissions, waste and ecological harm.

Shareholders own shares in a company and are a subset of stakeholders who care strongly about returns and firm value, and who can sell shares or the business to realise gains. Other stakeholders may have no ownership stake at all - staff, buyers, suppliers, the state, communities and the environment - yet they care about jobs, prices, payment, rules and local quality of life, and they can enable or block the firm even without shares.

Owners invest capital and seek a payoff for risk; successful trading can raise business value - often reflected in share prices for companies - which owners may realise by selling shares or the whole business, though profit need not be their only goal when solving customer problems and contributing to social welfare also matter. Managers and employees depend on the firm for income and often for identity, while the firm depends on them in return, and shared values and objectives support long-run success. Suppliers and customers create mutual dependency because quality, quantity and timing of inputs must match production needs while payment and future orders keep suppliers alive, and because customers need reliable offers while the firm needs their demand. Communities and government are affected through jobs, taxes, congestion and pollution, and environmental responsibility requires concrete results rather than greenwashing - claiming friendliness without proven action.

Interests conflict because stakeholders pull in different directions: faster delivery may raise owner returns and please customers yet harm residents and exhaust staff; lower prices delight buyers but squeeze supplier margins; stricter environmental filters protect rivers and may raise short-term costs. Good management makes conflicts visible and negotiates trade-offs instead of pretending one group is the only stakeholder.

At MeadowMill, the night-shift proposal touches family owners, plant managers, production staff, milk suppliers, supermarket buyers, municipal government, nearby residents and the river ecosystem. Owners and some buyers gain speed, residents lose quiet, staff face fatigue risk, and wastewater timing may worry environmental monitors, so decision quality depends on recognising the conflict early rather than denying that residents or staff are stakeholders. Conflicting interests are normal; classification skill is listing who is affected and what each group wants before choosing a path. If a city council cuts a supplier's night delivery permit to protect residents, nearby residents and the municipal government gain on quiet and regulation, while managers, owners and staff inside the firm are most likely pressured next by slower fulfilment and higher coordination cost. Equating "stakeholder" with "shareholder," assuming only people inside the firm count, or treating environmental claims as enough without activities and proven results, are the usual traps. Questions often mix ownership language with staff, suppliers, communities and environment, so ask who is affected or interested, who owns shares, whether the group is internal or external, and where aims clash. Legal structure and finance in later chapters sit beside stakeholder management as success factors, and Chapter 4 turns to ownership forms and finance - how shareholders, partners and sole owners differ legally - while this section keeps the wider interest map in view.

Chapter recap

- Businesses combine land, labour, capital and entrepreneurship/enterprise (plus knowledge and technology) in different mixes.

- Primary extracts, secondary manufactures, tertiary serves; developed economies are typically tertiary-heavy; GDP measures activity with well-known limits.
- Profit-oriented firms chase surplus for owners; NPOs chase mission while still needing revenue and reinvesting surplus.
- Size uses staff, turnover and balance-sheet total; course SME bands run micro -> small -> medium, with large above the ceilings.
- Reach runs local -> national -> international/multinational, with funding, logistics and border complexity rising along the way.
- Map stakeholders beyond shareholders; separate internal from external groups; expect and manage conflicting interests.

Chapter 4

Forms of business ownership and sources of finance

Before money moves, someone must answer a legal question: who is the firm? One owner with private assets at stake, several partners sharing risk, or a company that exists as its own legal person changes everything - who can be sued, who decides, and which doors to finance open. This chapter walks that path: ownership forms first, then the menu of equity and debt, then how to choose funding without mismatching purpose, cost, control, and gearing.

Learning objectives

1. Explain sole proprietorship: owner equals manager, easy setup, unlimited liability, and fragile continuity.
2. Distinguish general and limited partnerships, including unlimited liability for general partners and the role of a partnership agreement.
3. Describe corporations as separate legal persons with limited liability, shareholders versus directors, and Austrian AG/GmbH capital notes.
4. Compare ownership forms using an overview figure and a structured comparison table.
5. Classify sources of finance as equity or debt and as internal or external, and recognise common instruments.
6. Choose finance by weighing cost, risk, control, flexibility, purpose, and gearing (leverage).

4.1 Sole trader / sole proprietorship

One person, one shop. Mara opens a bicycle repair booth near a station. She registers as a sole trader, puts EUR 9,500 of her savings into tools and a workbench, and signs the rent herself. Customers deal with Mara - not with a company name that exists separately from her. When a supplier invoice is unpaid, the claim is against Mara personally.

Sole proprietorship (sole trader) A business owned by one person who also manages and runs it. The firm is not a separate legal entity: owner and business are, legally, the same person.

How sole trading works day to day is shaped by that identity of owner and firm. Setup is usually the easiest of the main forms, because there is typically no minimum share capital barrier of the AG/GmbH type, and the owner makes the key management decisions while bearing the risk. Profits are reported on the owner's personal income tax statement, and continuity is fragile: retirement, long illness, or death of the owner can interrupt or end the business unless someone else takes over. Staff can be hired, but the owner still owns the risk and the final decisions. Founders often choose this shell because it is simple and quick to establish, keeps full control of decisions, avoids sharing profits with co-owners, and taxes profits as personal income; what they must accept in return is unlimited liability with private assets at stake, finance limited by one person's wealth and credit, continuity risk if the owner cannot continue, and the concentration of all major decisions and risk in one person.

Unlimited liability The sole proprietor is liable for all debts and obligations of the business. If business assets are not enough, private assets can be claimed by creditors.

Finance for a sole trader leans heavily on the owner's own capacity. Personal savings and other outside money from investors or banks are external sources; once the shop earns a surplus that is left in the business rather than withdrawn, that retained profit becomes an internal source and carries no interest charge. Short-term tools such as a bank overdraft and trade credit are common, while longer bank loans may need collateral, often property as a mortgage. A common mistake is to think that "small business" automatically means "safe private assets": size does not create limited liability, and a tiny sole trader still faces unlimited liability. In exam stems, one owner who manages, is not a separate legal person, reports profits on personal tax, and puts private assets at risk points to sole trader with unlimited liability; continuity questions often test illness or retirement of that single owner. If Mara wanted a partner who invests money but never manages the shop, sole proprietorship would no longer be the right legal shell, because that structure admits only one owner - the investor role she has in mind belongs instead to partnership forms discussed next.

Worked: Mara's first-year money. At start-up she invests EUR 9,500 savings - external equity from the owner's point of view as funds put into the firm - and opens a current account with a EUR 2,500 overdraft limit. In operations she buys spare parts on 30-day trade credit, so unpaid purchases are short-term liabilities. Year 1 profit before drawings is EUR 22,000; she withdraws EUR 16,000 for living costs and leaves EUR 6,000 in the business, and that EUR 6,000 left in is internal finance (retained earnings) on which no interest is owed. If sales crash and the overdraft plus supplier bills cannot be cleared from business cash, creditors can pursue Mara's private assets under unlimited liability. Easy setup and full control therefore come with personal liability and finance that depends on one owner.

4.2 Partnership

Two names on the door. Leo and Nina want to run a landscape design practice together. Neither wants sole-trader loneliness: they want to split design and client work, pool savings, and share decisions. They must still answer who is liable if a project goes wrong and the bank is unpaid.

Partnership A business jointly founded by two or more persons. Partners set rights, responsibilities, and profit/loss sharing in a partnership agreement. In Austria, a general partnership of this type is called Offene Gesellschaft (OG).

Agreement first, then daily work. The partnership agreement typically settles ownership shares, profit and loss splits, roles, decision rules, dispute resolution, and terms of the partnership. In a general partnership, partners jointly own and manage and each has unlimited liability for the firm's debts, so a creditor may pursue any general partner for the full remaining debt, not only that partner's "share." Tasks can be specialised and ideas exchanged, which can improve decisions, and finance resembles sole trading but with more people who can contribute savings and supply collateral.

General partnership Partners jointly own and manage the business; each general partner has unlimited liability for the debts of the firm. Austrian label: Offene Gesellschaft (OG).

Limited partnership A partnership with at least one partner who is not involved in management and whose liability is limited to the amount contributed. Austrian label: Kommanditgesellschaft (KG). General (managing) partners still face unlimited liability.

The contrast between general partner and limited partner is therefore about management and liability together. A general partner is involved in management, faces unlimited liability, can bind the firm through management acts within the structure, and is typical in an OG as well as present in a KG as a managing partner. A limited partner is not involved in management, has liability capped at the contribution, provides capital while staying out of day-to-day running, and is the typical role in a KG alongside general partners. A limited partner must stay out of management if they want liability limited to their contribution, because stepping into management can undermine the limited status that the law attaches to that passive capital role. Another trap is treating the profit-sharing ratio as a liability cap: in a general partnership any general partner can be pursued for the whole remaining debt. Exam recognition for "two or more owners, agreement, unlimited for general partners, OG/KG" points to partnership; a silent capital provider with no management role and capped loss matches the limited partner / KG pattern. Partnerships remain unincorporated - like sole traders, they are not separate legal persons in the corporate sense you meet in 4.3 - and that is why unlimited liability for general partners matters so much.

Worked: Leo, Nina, and a bank claim. They form an OG. Leo contributes EUR 18,000; Nina contributes EUR 12,000. After salaries for work done, they share profits 55:45. The firm borrows EUR 35,000 to fit out a small studio. Business assets are later exhausted; EUR 14,000 of bank debt remains. The bank may claim the full EUR 14,000 from Leo or from Nina (or both), not merely 55% from Leo and 45% from Nina. Alternatively they form a KG: an investor, Priya, contributes EUR 20,000 as limited partner and takes no management role, so Priya's loss is normally capped at EUR 20,000 while Leo and Nina as general partners still face unlimited liability. Profit shares do not cap a general partner's liability; only a true limited-partner status limits loss to the contribution.

4.3 Corporations / companies

The firm that can sign its own name. Three engineers want to build sensor modules. They need serious capital, want investors who do not all run the plant, and do not want every supplier claim to reach their private homes. They look at forming a company - a legal person that can own assets, hire staff, sue and be sued in its own name.

Corporation (company) An incorporated business with separate legal personality. Ownership is divided into shares. Shareholders' liability is typically limited to the capital they invested. Owners need not manage; managers need not own shares.

Separate person, limited loss, directors in charge. As a legal person, the company can own land and property, hire people, close contracts, and litigate. Shareholders provide share capital; they are neither obliged nor automatically entitled to manage day to day. Shareholders elect a board of directors to make major decisions and represent owners. The top executive is often called CEO; other roles may include CFO, COO, CIO, or CMO. Setup is harder than for unincorporated forms, and capital rules may apply - but limited liability is the core protection for owners.

Limited liability Shareholders are usually liable only up to the money invested when buying shares. Creditors claim against the company, not automatically against shareholders' private assets.

Share capital Ownership capital divided into shares (stock). Buyers of shares become shareholders. Share capital raised at issue is long-term or permanent capital and is usually not redeemed by the company like a loan.

Share slice of capital

Value per share at issue = total share capital ÷ number of shares

Example course pattern: EUR 1,000,000 capital ÷ 100,000 shares = EUR 10 per share (0.001% of capital each).

Shares can be bought at initial issue or later from another shareholder. A stock exchange is a regulated market where listed shares and other securities (such as bonds) can be traded, and listing requires compliance with rules. An initial public offering (IPO) introduces shares at an issue price; afterwards prices move with demand and supply. Rising secondary-market prices enrich trading shareholders - they do not inject extra cash into the issuing company. Investors may buy shares to support a business they believe in; dividends are parts of profit paid to shareholders, with no automatic duty to pay every year; capital growth is the hope that share prices rise; common stock typically includes voting rights at the stockholders' meeting; and demand for shares reacts to firm prospects and to macro signals such as growth, interest rates, and inflation. A friend who says "my shares doubled on the exchange, so the company just raised twice the capital" confuses secondary trading with new corporate finance: the gain accrues among traders, not as fresh cash on the company's books. Two further confusions are worth avoiding: owning shares is not the same as managing, so shareholders must not be treated as directors by default; and secondary share-price gains must not be treated as new corporate finance.

Names differ by country: public limited company (PLC) in many English-speaking settings; Inc. or Ltd in others. In German-speaking countries the Aktiengesellschaft (AG) is the share corporation - Aktien means shares, Gesellschaft means corporation. In Austria, founding an AG requires minimum capital of about EUR 70,000. The European Company (SE) is a related form under EU law. Private limited forms keep legal personality and limited liability but usually do not offer shares to the general public on an exchange; examples include the US LLC, the UK private company limited by shares, and Austria/Germany's Gesellschaft mit beschränkter Haftung (GmbH). In Austria, GmbH minimum capital is about EUR 35,000 - lower than for an AG, still a barrier versus a partnership, and without the public flotation path of a listed AG. Large corporations can also issue bonds: packaged borrowing from many investor-creditors with agreed interest and repayment. Bond interest is often lower than a comparable bank loan for big issuers, which helps fund plants and infrastructure, yet bondholders remain creditors, not owners. Investor metrics on listed shares include price path, market capitalisation (shares outstanding x market price), dividend yield, and the price-earnings (P/E) ratio. Exam recognition for separate legal person plus limited liability plus shares plus directors elected by shareholders points to corporation; Austrian numbers to remember are AG ≈ EUR 70,000 minimum capital and GmbH ≈ EUR 35,000; and bondholder is not shareholder. Share capital

and bonds both appear again when you classify equity versus debt in 4.5 and when gearing rises in 4.6. In short, a corporation is a separate legal person; shareholders own while directors manage and the roles need not overlap; limited liability typically caps owners' loss at invested capital; Austrian AG $\approx$ EUR 70k and GmbH $\approx$ EUR 35k set minimum capital, with GmbH shares not for public exchange flotation like a listed AG; and IPO cash is company finance while later market trades are not.

Worked: issue price vs later market price. SensorCo issues 80,000 shares at EUR 20 each at IPO $\rightarrow$ share capital raised = EUR 1,600,000. Two years later the shares trade at EUR 31. Market cap = 80,000 $\times$ EUR 31 = EUR 2,480,000. The EUR 11 per-share rise goes to investors trading among themselves; SensorCo does not receive that secondary gain as new finance. Separately, SensorCo issues a EUR 900,000 bond at 3.5% to expand a clean room. Bond investors are creditors. If SensorCo is structured as an Austrian GmbH instead of floating publicly, founders still enjoy limited liability once capital rules are met, but shares are not sold to the general public on an exchange. Issue raises company cash; later price rises do not. Bonds add debt, not ownership.

4.4 Overview comparison of ownership forms

Three founders, one vehicle fleet. Jonas, Ela, and Rafi need about EUR 95,000 of vans for a cold-chain delivery start-up. They can go sole trader (one of them), form an OG, or meet capital rules for a GmbH. The legal shell changes liability, continuity, and who must manage.

Unincorporated vs incorporated Unincorporated businesses (sole trader, partnership) are not legal entities of their own. Incorporated businesses (corporations / limited companies) are legal persons. Among unincorporated firms, owners and managers are typically the same people; among incorporated firms, shareholders and directors need not be the same.

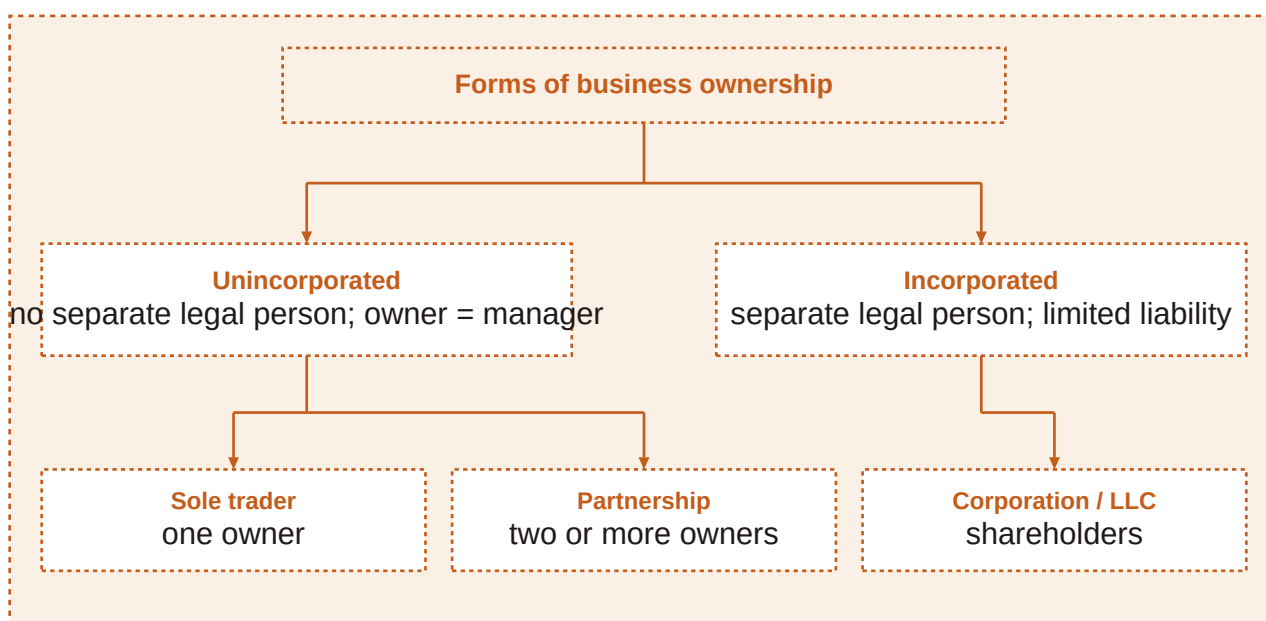


Figure 8. Forms of business ownership: unincorporated (sole trader; partnership) versus incorporated (corporations / limited liability companies), with ownership and management roles.

Reading the ownership map starts with legal personality. If none: one owner -> sole trader; two or more -> partnership (general and possibly limited partners). If separate legal person: shareholders provide capital; directors run the firm; liability of owners is typically limited to invested capital. Austrian AG/GmbH capital thresholds sit on the incorporated branch. When classifying in an exam, sort by legal personality first, then by number of owners, then by liability, and use the table mentally rather than inventing hybrid rules such as a "sole trader with limited liability." A related trap is assuming that "more owners" automatically means limited liability: an OG with three partners still exposes general partners to unlimited liability. If one founder's family home is their only major private asset, the liability row of the comparison is usually the decisive one.

Criterion	Sole trader	Partnership	Corporation / company
Legal personality	None - firm identical with owner	None - unincorporated	Separate legal person
Owners	One person	Two or more partners	Shareholders
Management	Owner manages	Partners manage (limited partners do not)	Board / directors; owners need not manage
Liability	Unlimited	Unlimited for general partners; limited for limited partners (to contribution)	Limited to capital invested in shares
Taxation of profits (course focus)	Personal income tax of owner	Typically personal tax on partners' shares	Corporate framework (company and/or shareholders)
Continuity	Fragile if owner retires or falls ill	Depends on agreement and partner succession	Stronger - legal person continues
Typical finance access	Owner savings, overdraft, trade credit, bank loans	Partners' capital, loans, overdraft, trade credit	Share capital, loans, bonds (larger firms); broader investor access
Setup / minimum capital	Easiest; usually none	Agreement required; usually none	Harder; AG $\approx$ EUR 70,000, GmbH $\approx$ EUR 35,000 (Austria)

Table 5. Comparison of sole trader, partnership, and corporation

Worked: cold-chain fleet options. Sole trader path: Jonas alone owns the vans, faces unlimited liability, funds mainly from personal savings plus a bank loan; Ela and Rafi are employees or contractors, not co-owners. OG path: all three manage, pool capital toward the EUR 95,000 fleet, share unlimited liability; a partnership agreement sets profit splits and decision rules. GmbH path: they meet Austrian minimum capital rules, take shares, enjoy limited liability to invested capital, and appoint managing directors (who may or may not be

shareholders). Trade-off: GmbH limits personal asset risk and clarifies continuity of the legal person; it costs more effort and capital than an OG or sole trader. Same vans, different answers to "who is liable?" and "who must manage?"

4.5 Sources of finance

Money has a passport. A boutique hotel renovation needs cash for a roof, seasonal linen stock, and a slow winter. Some euros will come from profits kept in the business; some from a new investor; some from the bank and suppliers. Each euro should be labelled: equity or debt? Internal or external? Short-term or long-term?

Equity finance Funds that belong to the owners of the business - for example share capital and retained earnings. Equity may be internal (retained earnings) or external (investor funds / newly issued share capital).

Debt finance Borrowed funds that create liabilities owed to creditors: bank overdrafts, trade credit, bank loans, bonds, and similar instruments. In the standard course overview, debt finance is external.

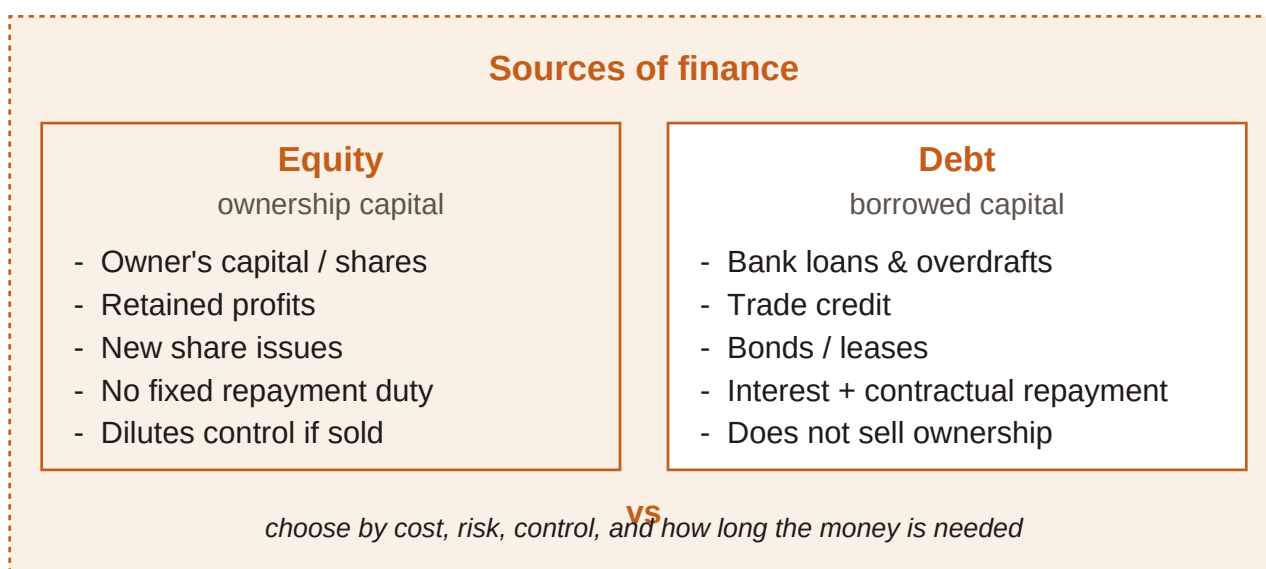


Figure 9. Sources of finance: equity (internal and external) versus debt (external), with common short-term and long-term instruments.

Equity finance (internal and external)	Debt finance (external)
Funds provided by investors (external)	Short-term credit: e.g. bank overdraft, trade credit, short-term (bank) loans
Share capital (external)	Long-term credit: e.g. bank loan, loan provided by owners, bonds
Retained earnings (internal)	

Table 6. Sources of finance (course overview)

How each common instrument works follows from those labels. Retained earnings are profit kept in the firm rather than distributed - internal equity with no interest charge. Share capital and investor funds are outside owners injecting ownership capital - external equity that is usually permanent and not repaid like a loan. A bank overdraft lets the firm draw beyond the account balance up to a limit, with interest only when overdrawn, as flexible short-term debt. Trade credit is delayed payment allowed by suppliers for purchases, also short-term debt - not "free equity" merely because no bank interest appears on day one. Bank loans are a fixed sum with agreed term and scheduled repayment, short- or long-term. Bonds are large-scale borrowing from many investors with contractual interest - long-term debt for bigger corporations. Owner loans remain debt if structured as lending that must be repaid, and are not the same as share capital. Leasing can finance use of a long-lived asset with regular payments instead of an outright purchase loan, which is useful when matching asset life. Venture-style investor equity fits the "funds provided by investors" external-equity cell when outsiders buy ownership stakes to fund growth. Two classification traps are especially common: calling share capital "internal" because shares sit on the company's books, when outside subscription makes share capital external equity; and treating trade credit as equity because suppliers are conversational "partners." Exam recognition: equity means owners' funds (internal retention or external shares/investors); debt means liabilities to creditors (always external in the course table); overdraft and trade credit are short-term debt; bonds and multi-year bank loans are long-term debt.

The balance sheet reveals which sources a business has used (see Chapter 6, Accounting). Large businesses are obliged to draw up such a balance sheet. Legal structure shapes the menu: sole traders and partnerships rely more on owner funds, overdrafts, trade credit, and loans; corporations add share issues and, for larger firms, bonds. Chapter 6 will show these sources on the balance sheet as equity and liabilities funding the assets side.

Worked: boutique hotel funding mix. Roof and bathrooms (multi-year assets): EUR 40,000 retained earnings (internal equity) + EUR 25,000 new share/investor capital (external equity) + EUR 45,000 five-year bank loan (long-term debt). Winter linen and cleaning stock: EUR 6,000 drawn on overdraft as needed (short-term debt) + EUR 9,000 of supplies on 45-day trade credit (short-term debt). Optional: lease a laundry machine for seven years instead of buying with a short overdraft - better duration match. Sale of an unused catering trolley for EUR 1,800 adds internal financing without interest. Labels: equity vs debt; internal vs external; short vs long - each line should have a clean passport. EUR 40k + EUR 25k equity and several debt lines can coexist; classification depends on ownership claim versus creditor claim, not on whether money feels "serious."

4.6 Choosing the source of finance

Cheap is not always right. A metal workshop can fund a EUR 140,000 laser cutter (twelve-year life) and EUR 18,000 of sheet metal for next month's jobs. The bank waves a revolving overdraft with a low headline rate this week. An equity partner offers capital that never needs monthly instalments but wants voting influence. Which criteria decide?

Finance choice criteria When several sources are available, selection usually weighs costs, the intended use (purpose) of the funds, and the firm's current financial situation - especially how much loan capital it already carries. Risk, control, and flexibility complete the practical

checklist.

How the criteria bite. Cost covers interest on loans and credit and the administration costs of issuing shares or bonds. Purpose and matching mean that capital expenditure on assets used for years needs long-term finance, while revenue expenditure such as materials can use short-term sources. Risk rises because debt must be repaid - too much raises insolvency risk. Control matters because new shareholders or partners may gain influence, whereas pure debt usually leaves ownership percentages unchanged but adds creditor pressure. Flexibility differs: overdrafts can be drawn and repaid within a limit, while committed long loans and covenants are less flexible. Gearing (leverage) asks how high loan capital already stands relative to equity: a high-gearred firm can find lenders reluctant, face higher interest, or be asked for collateral, which pushes it toward retained earnings or new equity investors instead of yet another loan. A firm might therefore reject the lowest advertised interest rate this month if the source mismatches purpose and duration, dilutes control unacceptably, reduces flexibility too far, or piles debt onto an already high-gearred balance sheet.

High gearing (high leverage) A high proportion of loan capital relative to equity. It raises insolvency risk and may make further credit harder or more expensive to obtain.

Gearing intuition

High gearing $\approx$ large loan capital compared with equity capital

Course focus is the idea and consequences, not a single mandatory exam formula. Example: if loans already equal about 75% of capital employed, adding a large new loan is often difficult or punitive.

Putting the checklist into questions makes the trade-offs concrete. On cost, ask what interest or issue and administration costs apply, and compare loan interest with share or bond issuance costs. On purpose, ask whether the need is capital expenditure or revenue expenditure, matching long assets to long finance and materials to short finance. On risk, ask whether the money must be repaid on a fixed schedule, remembering that heavy debt raises insolvency risk if cash dips. On control, ask whether new owners gain votes or vetoes - equity can dilute influence while debt usually does not. On flexibility, ask whether the firm can draw, repay, or refinance easily - overdrafts are flexible, locked loans less so. On gearing, ask how high loan capital already is; when geared high, prefer retention or investors over more debt. Exam stems that mention capital versus revenue expenditure test matching; stems that mention reluctance of lenders, collateral, or insolvency risk test high gearing; stems that mention voting or ownership percentages test the control trade-off of equity. The traps to avoid are choosing finance by interest rate alone, funding a long-life machine with an overdraft, and piling on debt when already high geared because "growth needs money." In practice, judge sources by cost, purpose, risk, control, flexibility, and gearing rather than by a single number; match long assets to long finance and short needs to short finance; recognise that high gearing raises repayment risk and can block or price up new debt; and when geared high, prefer internal funds or equity investors over another heavy loan. The legal form in 4.1-4.4 shapes which sources in 4.5 are even on the menu; 4.6 decides which menu item to pick under those criteria.

Worked: laser cutter vs sheet metal. Laser cutter (EUR 140,000, ~12-year life): funding via revolving overdraft mismatches duration - the bank could demand repayment long before the machine pays for itself. Better: long-term loan, lease, or equity/retained funds. Sheet metal (EUR 18,000 for next month): trade credit or a short-term facility matches the short cycle. Cost check: a bond or share issue has admin costs; a loan has interest - compare total cost, not only the poster rate. Control check: an equity partner funding the cutter may want board influence; a loan leaves share percentages intact but adds repayment risk. Gearing check: if loans already $\approx 75\%$ of capital employed, lenders may refuse or charge more / demand collateral. Retained profit or an equity investor may be the feasible path. Match purpose and duration first; then price the cost, risk, control, flexibility, and gearing consequences.

Chapter recap

- Sole traders are owner-managed, easy to set up, face unlimited liability, and have fragile continuity.
- Partnerships use an agreement; general partners have unlimited liability (OG); limited partners (KG) cap loss at their contribution and stay out of management.
- Corporations are separate legal persons with limited liability; shareholders own, directors manage; Austrian AG $\approx$ EUR 70k and GmbH $\approx$ EUR 35k minimum capital.
- Compare forms with the ownership-overview figure and the criterion table (personality, liability, management, continuity, finance, setup).
- Finance splits into equity vs debt and internal vs external; know share capital, retained earnings, overdraft, trade credit, loans, bonds, and related tools.
- Choose funding by cost, risk, control, flexibility, purpose (matching), and gearing - cheap debt is not always the right debt.

Chapter 5

Marketing

A workshop can craft the finest desk lamp in the city and still fail if nobody needs light in that form, at that price, in that place. Marketing is the disciplined work of finding who those people are, shaping what is offered to them, and making the exchange happen - not only advertising, but research, design choices, pricing logic, distribution routes, and messages that stay consistent with one another.

This chapter walks that path from product idea to marketing mix. You will learn how firms set marketing objectives, choose between product and market orientation, research customers, segment markets, and blend the four Ps, including life-cycle thinking and the Boston Consulting Group matrix. By the end, you should be able to explain why a strong product still needs a market, and how each marketing tool supports the others.

Learning objectives

1. Define a product in marketing terms and distinguish goods, services, product lines, product mix, branding, USP and differentiation.
2. State typical marketing objectives and explain how they connect (satisfaction, loyalty, awareness, sales, market share, profitability).
3. Contrast product orientation with market orientation and recognise the role of customer relationship management.
4. Discuss responsibility and sustainability pressures on marketing practice for firms and consumers.
5. Distinguish primary from secondary research and qualitative from quantitative approaches; calculate absolute and relative market share.
6. Apply segmentation bases and compare undifferentiated, differentiated and concentrated (niche) targeting.
7. Build a coherent marketing mix (product, price, place, promotion) and place offerings on the product life cycle and BCG matrix.

5.1 Product: goods, services, lines, brands and differentiation

Same object, different product story. River & Reed Workshop sells modular oak desk lamps: a co-working landlord buys twenty for shared desks and books a lighting consultation so staff know how to adjust glare, while a student buys one lamp for a rented room and never needs the consultation. Physically the lamp looks identical, yet in marketing the exchanges are not the same product situation - one sits in a business-to-business relationship with an attached service, the other is a business-to-consumer purchase of a good alone.

Product (marketing) Every good and/or service that can be exchanged to fulfil customers' wishes and needs. Tangible items and intangible support both count when they can be offered in exchange.

Customers may be other businesses or private households. Goods and services sold from one business to another are producer products (business-to-business, B2B); the same

categories sold to consumers are consumer products (business-to-consumer, B2C). Buyer and user need not be the same person: a parent may pay while a teenager uses the product, or a facilities manager may purchase while employees use the lamps daily. Value is not only the core function - light on a desk - because extra features such as dimmers or USB ports, image such as craft reputation, and the joy or reassurance of use all shape why someone chooses one offering over another. Marketing designs and communicates that whole package, not only the physical object.

Product line A set of closely related products - for example several desk-lamp models that differ in size, wood finish or power options but belong to the same family.

Product mix The full range of product lines a business offers. Width increases when new lines are added (lamps plus wall sconces). Depth increases when more variants sit inside one line (more finishes of the same lamp).

Portfolio shape can move in several directions. A firm may specialise by competing with one line and knowing that line deeply, or diversify by adding lines (mix extension) to reach more needs or spread risk. Line extension adds variants inside an existing line - new sizes or colours - while a relaunch uses minor changes such as packaging or colour to refresh interest. Contraction drops weak products or whole lines when relaunch no longer pays.

Brand A name, word(s), symbol and/or sign that distinguishes a product or business from others. Brands support recognition, signal a promised quality level, and help build loyalty when experience matches the promise.

Unique selling proposition (USP) A distinctive benefit - real or clearly perceived - that sets an offering apart from similar products. Differentiation may rest on a product characteristic or on how the offer is promoted and understood.

Branding and USP reinforce each other: a brand is the identity vehicle, and a USP is the reason to prefer that identity. Without a believable difference, branding becomes empty decoration; without a recognisable brand, a genuine difference is harder to remember and harder to charge for. Product differentiation aims to make the offer special, unique, or better for this customer than close substitutes.

Goods and services often sit inside the same firm. Goods are physical units that can be stored and shipped, ownership typically transfers to the buyer, and quality can be inspected before use. Services such as lighting consultation and installation advice are produced and consumed largely together, and quality is often judged during or after delivery. Classify River & Reed's offers accordingly: an oak desk lamp sold to a household is a consumer product (B2C good); the same lamp sold to a co-working firm is a producer product (B2B good); an on-site lighting consultation is a service product that can be B2B or B2C; three heights of the same oak lamp form one product line (depth); adding a wall-sconce line is mix extension (greater mix width); and a USP claim such as "tools-free brightness change in under ten seconds" differentiates on a usable feature, not only on slogan tone. Marketing treats all exchangeable goods and services as products; lines and mix describe portfolio shape; brand and USP explain preference. A common mistake is treating "product" as only a physical object - repair contracts, tutoring hours and software licences are products if they are exchanged to meet needs - and when a stem switches the same device from office use to

home use, the classification flips from producer (B2B) to consumer (B2C) even though the item is unchanged. USP also requires a distinctive benefit or perception, not mere advertising noise.

5.2 Marketing objectives

Five targets on one whiteboard. Before PeakMeal, a meal-prep start-up, spends on ads, the founders write five annual targets: raise average satisfaction scores after delivery, lift brand awareness among night-shift hospital staff, grow unit sales by 15%, raise absolute market share in the city meal-prep niche from 6% to 9%, and keep contribution high enough for a thin operating profit. The list looks mixed - soft feelings and hard euros - but each target shapes different marketing choices.

Marketing objectives Specific aims that guide market analysis and marketing action. Typical aims include customer satisfaction and loyalty, awareness, a unique selling proposition, sales volume or value, market share, and profitability. Objectives often reinforce each other rather than standing alone.

The main objectives link more tightly than a checklist suggests. Dissatisfied customers rarely buy again, so satisfaction feeds loyalty and future sales; awareness matters because unknown offers are not chosen; a USP reduces pure price comparison and supports preference; market share shows relative competitive position; sales generate the revenue needed to cover costs; and profitability reimburses owners and funds reinvestment - higher sales help, but only within cost and price limits. Customer satisfaction asks whether the offer met or beat expectations, because unsatisfied buyers leave and satisfied buyers often return. Loyalty asks whether they will choose us again and recommend us, which lowers acquisition pressure. Awareness asks whether target customers know we exist and what we offer, since choice requires recognition in the relevant set. USP and differentiation ask why us rather than a close substitute, supporting preference beyond lowest price. Sales ask how many units or how much revenue, because cash must flow in to cover costs and grow. Market share asks what fraction of the defined market is ours, signalling relative competitive strength. Profitability asks whether revenues cover costs with a surplus, rewarding capital and funding reinvestment.

Translate PeakMeal's whiteboard into actions. Satisfaction means shorten delivery windows and fix cold-meal complaints within 24 hours. Awareness means partner with two hospital noticeboards and a night-shift podcast, not random city-wide billboards. The USP - "balanced meals that survive a 12-hour shift without reheating" - is a concrete difference. Sales and share track units in the defined city meal-prep market, not all food retail. Profitability means avoiding deep discounts that raise share but destroy contribution. Objectives only work when the market is defined and tools (product, price, place, promotion) are aligned with those aims. If awareness doubles while satisfaction collapses, short-term sales and recognition may rise while loyalty, repeat sales and next-quarter profitability are likely to fall. Market share is not automatically identical to profit, because share can rise through unsustainable discounts; a USP is not a slogan with no product or perception difference behind it. Market share objectives need clear market measurement (section 5.5), while USP and branding connect back to product differentiation (section 5.1) and forward into the marketing mix (section 5.7).

5.3 Product orientation versus market orientation

Two clockmakers, two starting points. Atlas Gears spends eighteen months perfecting a mechanical desk clock with a sapphire face, then asks a salesperson how to push it into shops. North Harbor Time first interviews ferry crews and harbour offices about what fails in salty air, then designs a sealed quartz clock with oversized numerals for glove use. Both want reliable sales and a fair margin, yet their starting questions differ: "How do we sell what we built?" versus "What should we build because customers need it?"

Product orientation An approach that focuses first on production and product features, then seeks to sell what has been made. Success is expected mainly from the quality and specifications of the product itself.

Market orientation An approach that starts from customers' needs and wants, then designs and adapts offerings to match them. Success depends on reading and responding to the market.

Product orientation focuses on production and features first, then arranges sales and advertising, under the belief that a strong product largely sells itself. Market orientation focuses on needs and wants first, then produces what customers require, with the advantage of earlier response to market change. Both orientations may pursue identical marketing objectives such as sales, share and satisfaction; the product-oriented firm leans on feature quality to win, while the market-oriented firm leans on continuous customer insight. Suitability depends on the product and on how many competitors exist, but neglecting customer expectations is risky in either case. Identical objectives do not prove identical orientation, because the sequence - build-then-sell versus research-then-build - can still differ.

Customer relationship management (CRM) Practices that aim at a long-term relationship with customers by keeping contact data and purchase history to send relevant information, offers and reminders - so customers return. Sensitive handling of personal data is indispensable.

Loyalty cards and personal accounts often mean customers willingly trade anonymity for discounts and tailored offers. Firms then observe buying behaviour and personalise communication. That power helps market-oriented selling, but it also raises responsibility for privacy and consent; CRM is not "free data" with no sensitivity duties. Atlas Gears can reorient without abandoning craft by interviewing harbour and outdoor workers about fog, gloves and vibration, keeping the mechanical line for collectors as a product-strength niche, launching a sealed, high-contrast model for working docks as a market-led line, and using optional CRM emails only with clear opt-in for service reminders. Orientation is a starting logic, not a ban on engineering excellence: market orientation still needs real product competence, and neither approach should ignore the market entirely. Even a pharmaceutical firm developing a new active ingredient may start closer to product orientation at the research stage, yet it must still not neglect customer needs, safety expectations and market fit before launch.

5.4 Responsibility and sustainability in marketing

The jacket worn three times. A fashion pop-up pushes "drop" launches every fortnight. Mira buys a neon windbreaker for a festival, wears it three times, and bins it when the zipper fails. The brand's marketing created urgency she did not plan for. Across town, Mend & Wear runs a repair desk, sells durable jackets with spare zippers, and rents formal coats for weddings. Same clothing market; different stance on how far marketing should push consumption.

Responsible and sustainable marketing Practice that recognises marketing's power to shape wants, not only to respond to them, and that seeks production and consumption patterns with lower waste, longer use, and fairer treatment of customers and society.

Marketing can inflate want. Continuous novelty and persuasive promotion can create desires people would not otherwise have had, encourage spending beyond intention or means, and speed up obsolescence of still-usable goods. Critics focus on these effects; defenders emphasise information and matching real needs. Either way, both businesses and consumers influence how sustainable the outcome is, so blaming only firms or only consumers misses the shared role. Firms can design for durability, repairability and honest claims, and avoid deceptive urgency; consumers can question whether a purchase will be used and prefer reuse, repair and rental where sensible. The shared aim is to reduce wasteful churn without denying genuine needs. Repair-and-reuse models keep devices and clothes in use longer; clothing rental for rare events can replace one-off cheap purchases; swap and second-hand circuits also shift status away from constant newness. Marketing can promote these patterns instead of only "upgrade now." A green colour in an advert is not sustainable marketing if product and process substance are missing.

Mend & Wear can rewrite a campaign season by replacing "new drop every Friday" with "repair week" and free zipper clinics, advertising rental for one-off events instead of disposable formalwear, publishing battery-health or fabric-care facts rather than vague "eco" labels, and training staff to suggest mend-first when a product still has life. Responsibility shows up in product design, claims, pricing of repair versus replacement, and promotion tone - not in a slogan alone. A "buy one, get one free" deal for single-use plastic cutlery sits poorly beside a sustainability claim unless the offer itself changes toward durable, reusable or genuinely lower-waste alternatives. Sustainability choices feed the product P through durable design, promotion ethics, and sometimes pricing of repair services; they also interact with market orientation when firms listen to customers who already demand lower waste.

5.5 Market research

Guesswork is expensive. GlassHarbor Cycles wants to know whether commuters will pay EUR 45/month for a battery-swap add-on. One founder "just knows" they will; another insists on evidence through street interviews, a short online questionnaire, and city mobility statistics already published by the transport agency. Research will not remove all risk, but it replaces pure hope with structured information about customers, rivals and the industry.

Market research The collection and interpretation of information about existing and prospective customers, potential buyers, competition and the industry, used to support marketing decisions.

Primary market research Original data gathered for the firm's own questions - for example surveys, interviews, observation or experiments, sometimes via a research institute. Tailored

and specific, but often costly.

Secondary market research Use of existing data collected by others - government statistics, trade association reports, published studies. Usually cheaper and faster, but more general and not built for the firm's exact problem.

Qualitative research explores motives, language and experiences in depth through interviews, focus groups and open observation notes; samples are smaller, meaning is rich, and statistical generalisation is limited. Quantitative research measures amounts, frequencies and relationships with numbers through structured questionnaires, counts and experiments with metrics; larger samples support stronger estimates of shares and averages. The two pairs work together: primary versus secondary answers who collected the data and for what, while qualitative versus quantitative answers what kind of insight. A firm might use secondary city cycling counts (quantitative secondary), then primary interviews about battery anxiety (qualitative primary), then a priced survey of 400 riders (quantitative primary). Primary research is tailored to the firm's exact questions - useful for new product concept tests and satisfaction drivers - but costs time, money and design skill. Secondary research is faster and often cheaper or free - useful for market sizing and industry growth context - yet it may be generic, outdated or mismatched. A questionnaire the firm designs itself is primary, not secondary, research.

Customer analysis often asks who buys, what they do with the product, where they buy, when they buy, and why they choose one option over another. Buyer and user may differ; influencers such as children may shape a purchase paid for by someone else. Timing patterns reveal seasonality and opportunities for price differentiation across the year. A free government mobility report can be useful for sizing and context yet still insufficient before launching the EUR 45 battery-swap plan, because it will not answer willingness to pay or the firm's exact concept questions.

Market size (market volume) Total sales of a product by all businesses in a defined market, expressed as value (for example euros) or quantity (units).

Market share The proportion of a defined market held by a business, product or brand. Absolute share compares own sales to market volume; relative share compares own share to the largest competitor's share.

Absolute market share

Absolute market share = sales of one business (or brand) / market volume

Sales and market volume must use the same units (euros or units) and the same market definition.

Relative market share

Relative market share = own market share / largest competitor's market share

A value of 1 means parity with the leader; below 1 means smaller than the leader.

Market potential can exceed current market volume if unrealised customers remain. A firm's sales potential can exceed current sales if it can win rivals' customers or claim part of that

unrealised potential. Absolute share informs managers and investors; relative share adds competitive context the absolute figure alone hides. For GlassHarbor's battery-swap plan, define the market as city e-bike accessory subscriptions this year at EUR 800,000, with GlassHarbor subscription sales of EUR 96,000. Absolute share = $96,000 / 800,000 = 0.12 = 12\%$. If the largest rival holds 30% absolute share, relative share = $12\% / 30\% = 0.4$. If research estimates market potential at EUR 1,000,000, about EUR 200,000 of demand is still unrealised industry-wide. GlassHarbor is meaningful but far from leadership; relative share 0.4 makes that gap explicit. Do not mix value share with unit share without saying so, and do not report absolute share without defining the market boundaries.

5.6 Segmentation and targeting strategies

Not every rider wants the same bike. GlassHarbor's research shows four different "ideal bikes" living in one city: students wanting cheap short hops, nurses wanting reliable night commuting, tourists wanting day rentals near the harbour, and cargo users needing stable racks. Treating all four as one average customer would produce a bike that delights nobody. Segmentation starts by grouping similar customers; targeting chooses whom to serve; positioning shapes what those groups believe the offer stands for.

Market segmentation Dividing a market into relatively homogeneous subgroups of customers who share relevant characteristics for marketing decisions.

Targeting Evaluating segment attractiveness and selecting one or more segments to serve with a tailored strategy.

Positioning Creating a clear image or identity for the product in the minds of the chosen target market(s) so the offer fits their demands.

Segmentation is useful when groups are measurable in size and purchasing power, profitable, accessible through communication and distribution channels, and durable enough that they do not dissolve too quickly. Without those conditions, fine slicing wastes resources - a small customer group is not a valid segment if it cannot be measured, reached or served profitably.

Base	Example variables	What it helps decide
Geographic	City, region, country; urban vs rural	Coverage area, logistics, climate or language fit
Demographic	Age, gender, education, income, employment status	Purchasing power and life-stage offers
Psychographic	Lifestyle, values, attitudes (e.g. environmental concern)	Message tone and brand meaning
Behavioural	Usage intensity, occasions, loyalty, benefits sought	Service level, pricing, loyalty programmes

Table 7. Common segmentation bases

Target market A group of people or businesses towards whom a firm markets goods, services or ideas with a strategy designed to satisfy their specific needs and preferences.

Targeting strategies differ in how many segments they serve and how much the offer varies. Course language often says mass, segment and niche marketing; the same logic appears as undifferentiated, differentiated and concentrated (niche) targeting. Undifferentiated or mass marketing uses one offer for the whole market with the same promotion style for almost all, which can bring economies of scale but less flexibility, and is common for widely used staples such as soap or basic pens. Differentiated or segment marketing designs different offers for several segments and focuses resources where strategic fit is strongest, trading better fit against higher complexity and cost. Concentrated or niche marketing deepens focus on a narrow subgroup, trading depth and expertise against limited volume; small firms often niche, and specialists can lead despite size. Niche marketing is deliberate focus, not randomly ignoring research. Mass marketing of laundry detergent can still make sense because needs are widely shared and scale efficiencies matter, while mass marketing of specialised medical bikes usually does not, because needs are narrow and fit matters more than volume.

GlassHarbor chooses a niche inside a segment. Geographic focus is one coastal city and inner suburbs; demographic focus includes employed adults and shift workers; psychographic preference leans toward lower waste and dislike of unused ownership; behavioural focus is regular commuting rather than weekend hobby riding. The segment chosen is shift-working commuters with limited storage at home; the niche inside it is night-shift staff needing lit routes and battery-swap certainty; the positioning phrase is "get to the ward on time - battery panic optional." Concentrated targeting lets a small firm win relevance without matching mass producers on volume. Targeting is only complete when the marketing mix (section 5.7) delivers a consistent product, price, place and promotion for the chosen segment.

5.7 The marketing mix, life cycle and BCG matrix

Four dials, one sound. North Harbor Time's sealed dock clock is almost ready. If the price screams luxury while the sales channel is a discount warehouse, customers will not know what to believe; if promotion promises glove-friendly numerals but the product ships tiny markings, trust collapses. The marketing mix is the craft of setting product, price, place and promotion so they play in tune for the target market.

Marketing mix A harmonised blend of marketing tools that best meets the needs and wants of customers in the targeted market. Classically organised as the four Ps: product, price, place and promotion.

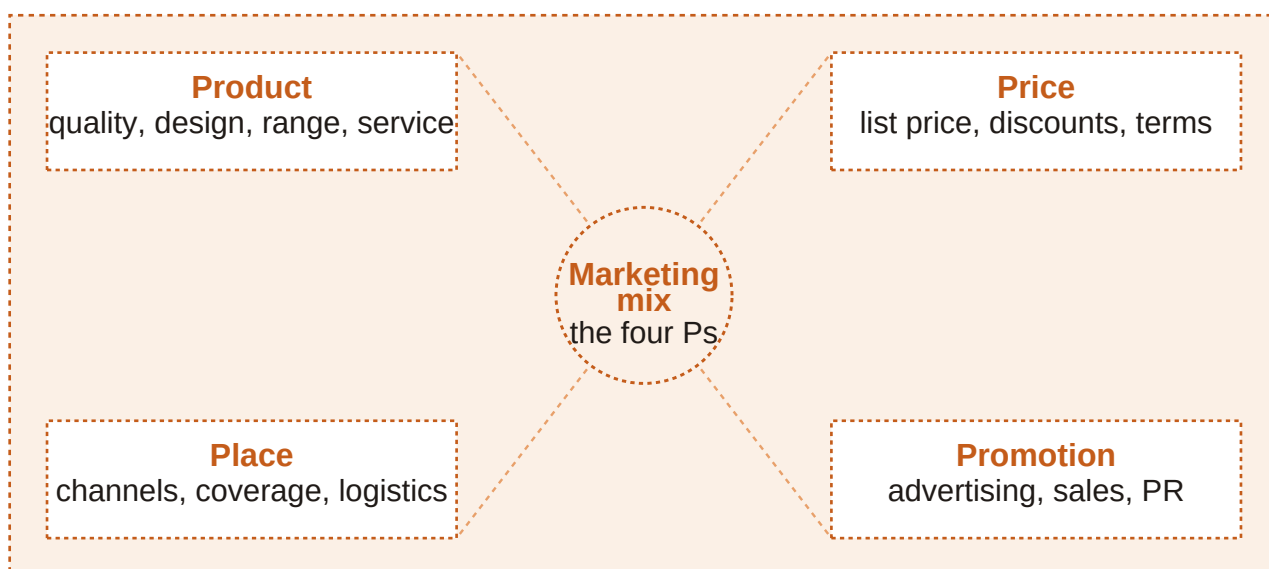


Figure 10. The four elements of the marketing mix: product, price, place and promotion - decisions that must stay consistent with one another and with market research.

The basic idea is simple to state and hard to execute: provide a suitable product, at a price customers will accept, available in a convenient place, supported by a clear promotional message. Market research should inform all four dials. Memorising the four Ps as a shopping list without harmonisation misses the point of the mix.

Product (as a mix element) All goods and services offered - the heart of the mix. Includes line and mix decisions, branding, quality level, features, packaging and supporting services.

Most firms offer more than one item. Similar items form a product line; several lines form the product mix. Brands distinguish offerings and support USP, recognition and loyalty; familiar brands also reassure travellers who meet the same marks abroad. Mix strategy may specialise on one line or diversify across more lines. Relaunches refresh; line extensions deepen; mix extensions widen; weak items may be eliminated.

Product life cycle A theoretical model of stages in a product's market life that differ in sales and profit: introduction, growth, maturity and decline (after development costs before launch).

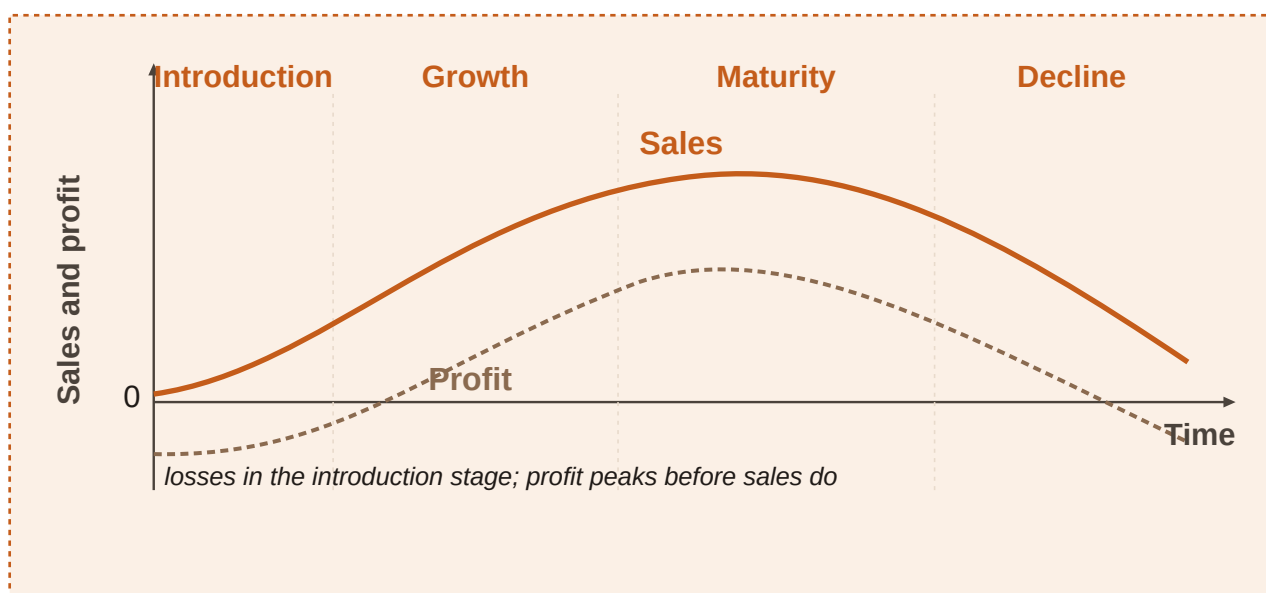


Figure 11. Product life cycle: introduction, growth, maturity and decline - sales and profit follow different paths across stages.

Before launch there are development costs and no sales. Introduction often starts in loss because promotion is heavy and prices may be set to attract trial among innovators and early adopters, with emphasis on market development. Growth brings faster sales into a widening mass market; average costs may fall with scale; profits usually appear and strengthen under a market-penetration emphasis. Maturity peaks sales in the mass market while growth slows and competition intensifies, so strategy becomes more defensive and profit peaks then comes under pressure. Decline sees falling sales and profits among loyal remainers, so efficiency or exit becomes central. Stage length varies: some staples linger in maturity for years; fads may fade within months.

Stage	Typical strategy emphasis	Customers	Sales	Profit
Introduction	Market development / trial	Innovators and early adopters	Low, then rising	None (often loss)
Growth	Market penetration	Widening mass market	Rapid growth	Strong, rising toward peak
Maturity	Defensive positioning	Mass market	Peak, slow growth	Peak then under pressure
Decline	Efficiency or exit	Loyal remainers	Declining	Low or none

Table 8. Product life cycle characteristics (course-style summary)

Boston Consulting Group (BCG) matrix A portfolio map of products by market growth (high/low) and relative market share (high/low), classifying them as stars, cash cows, question marks or dogs (poor dogs).

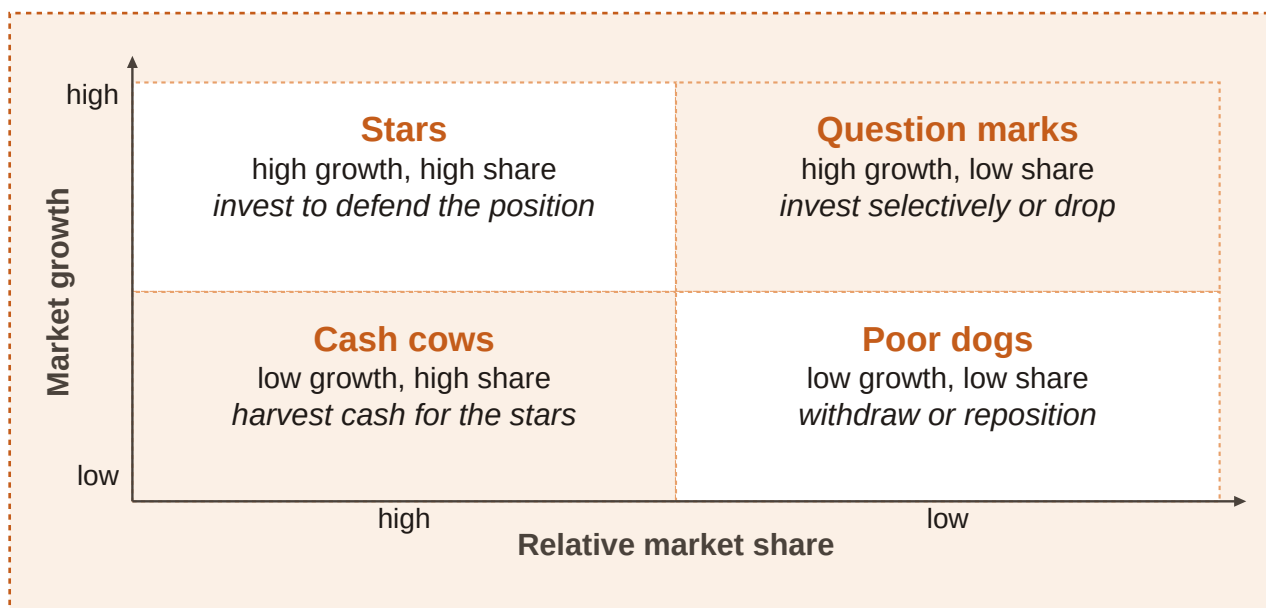


Figure 12. Boston Consulting Group matrix: stars and question marks in high-growth markets; cash cows and dogs in low-growth markets - split by high versus low relative market share.

Stars sit in high-growth markets with high relative share: they are valuable yet still need investment in promotion and capacity. Cash cows sit in low-growth markets with high relative share and generate cash with lower investment needs - so a high-share product in a low-growth market is a cash cow, not a star. Question marks sit in high-growth markets with low relative share and need decisions: invest to build share or withdraw. Dogs, sometimes called poor dogs, sit in low-growth markets with low relative share and have limited future, making them candidates for harvest or exit. A healthy portfolio blends stars, cash cows and selected question marks without too many dogs; cash-cow revenues can fund stars that may become future cows and promising question marks. Life-cycle stage and BCG cell often move together: introduction may look like a question mark, growth success can create a star, mature leaders become cash cows, and decline risks dog status. Extension strategies try to delay decline - for example by changing the product mix or entering new markets. Ansoff's product-market matrix organises options: market penetration (existing product, existing market - safest), product development (new product, existing market), market development (existing product, new market), and diversification (new product and new market - riskiest, but can spread risk long term).

Price The amount customers pay. Hard to reset casually once a strategy is established. Influenced by costs, competitors' prices and demand (willingness to pay). Lowest price is not automatically best.

Cost-based pricing seeks to cover costs and allow a profit margin; competition makes it hard to charge more without superiority or clear difference; demand through high willingness to pay supports higher prices. In the long run, total costs - fixed plus variable - must be covered. Variable costs rise with output; fixed costs such as rent and insurance do not vary directly with output in the short run. Break-even is where revenues equal total costs. Contribution per unit

is selling price minus variable cost per unit; it helps cover fixed costs and then profit.

Contribution and break-even

Contribution per unit = selling price - variable cost per unit

Break-even units $\approx$ fixed costs / contribution per unit

From the next whole unit above break-even, the simplified model shows profit (other factors held aside).

Variable-cost-plus, or distribution, pricing adds a markup to variable cost - common when bidding or launching under competitive pressure. The short-run absolute lower limit is variable cost: any price above it still contributes to fixed costs. Before changing prices, consider price elasticity of demand: elastic if the percentage quantity change exceeds the percentage price change; inelastic if quantity responds less. Few substitutes and necessities lean inelastic; luxuries and easy substitutes lean elastic. Short-run and long-run elasticity can differ as habits and substitutes adjust. Break-even is a planning model, not a guarantee of cash in the bank regardless of credit and stock timing. For a sealed dock clock with variable cost EUR 54 and planned selling price EUR 90, contribution = $90 - 54 = \text{EUR } 36$. With fixed costs of EUR 72,000 for the line this period, break-even units $\approx 72,000 / 36 = 2,000$ clocks. A bulk buyer offers EUR 70 each for 300 clocks: contribution = $70 - 54 = \text{EUR } 16$ each, or EUR 4,800 toward fixed costs. Accept if those 300 would otherwise sit unsold; decline if they surely sell at EUR 90 elsewhere. Markup targets and short-run acceptance decisions both rest on contribution logic, not on vanity list prices alone.

Place (distribution) How and where the product is made available. A strong product fails if customers cannot find it where they expect it.

Place can run through distributors, wholesalers who sell on to businesses, and retailers who sell on to consumers; through agents or brokers, especially in B2B; through online selling and telemarketing; and through formats such as vending machines and kiosks. Channel partners bridge factory and final buyer with logistics, local information and advertising support. Beverage makers, for example, rarely sell every bottle direct to households nationwide; they need partners for reach.

Promotion Activities that inform potential customers and the public about the business, its products and their benefits.

Promotion tools include advertising on television, radio, online, social media, print, outdoor and transit; direct mailing; personal selling through salespeople; sales promotions; sponsorship of events, people or organisations; and public relations for a favourable image and stakeholder relationships. Small budgets still have options: website, local ads, social presence, leaflets and word-of-mouth from satisfied customers. Promotion must match the target segment and stay consistent with product quality cues, price position and place.

A harmonised mix sketch for North Harbor's dock clock keeps the dials aligned. Product is sealed quartz with oversized numerals and a glove-friendly crown, matching harbour-worker needs from research. Price is mid-premium - above toy clocks and below luxury mechanicals - signalling durability without claiming jewellery status. Place is marine suppliers, selected

hardware stores and a direct web shop, where dock staff already buy kit. Promotion uses demo videos with gloves on, a harbour newsletter and a trade stall, showing the USP in use rather than abstract luxury imagery. If North Harbor cuts price by 40% to chase volume while keeping "premium sealed instrument" ads, customers will notice the price-position conflict first. The mix implements objectives (5.2) for chosen targets (5.6) using research (5.5). Orientation (5.3) shapes whether the product P starts from features or from stated needs. Responsibility (5.4) constrains claims and design inside the same mix.

Chapter recap

- A marketing product is any exchangeable good or service that fulfils wishes and needs; portfolios are shaped as lines and mixes, while brands and USPs drive differentiation.
- Marketing objectives - satisfaction, loyalty, awareness, USP, sales, market share, profitability - should be explicit and mutually consistent.
- Product orientation builds then sells; market orientation researches then builds; CRM extends relationships with careful data use.
- Responsible marketing faces the risk of inflating wants and supports more sustainable production and consumption patterns such as repair, reuse and rental.
- Market research combines primary/secondary sources and qualitative/quantitative methods; absolute and relative share quantify competitive position.
- Segmentation uses geographic, demographic, psychographic and behavioural bases; targeting may be undifferentiated (mass), differentiated (segment) or concentrated (niche).
- The marketing mix harmonises product, price, place and promotion; life-cycle stages and the BCG matrix guide how products are managed over time.

Chapter 6

Accounting - keeping record of business transactions

Every sale, purchase, wage payment and loan leaves a trail, and accounting is the craft of turning that trail into statements you can actually read - what the firm owns, what it owes, whether it earned a profit, and whether cash really moved. This chapter builds that picture step by step, from the balance-sheet identity through the income and cash-flow statements, and then shows how ratios turn the same numbers into answers about liquidity, profitability, efficiency and gearing.

Learning objectives

1. Construct and interpret a balance sheet using $\text{assets} = \text{liabilities} + \text{equity}$, and classify fixed (non-current) versus current items.
2. Explain the roles of the income statement and the cash flow statement, and calculate straight-line depreciation.
3. Distinguish expenditures from expenses, and read COS/COGS, gross profit, EBITDA and EBIT in a structured income statement.
4. Contrast financial accounting with management accounting and identify who uses accounts.
5. Analyse liquidity, profitability, efficiency and gearing by moving from intuition to formula to interpretation of the same firm.

6.1 The balance sheet - what the firm owns and how it is financed

Opening day at Northline Bike Workshop finds Mira and Jonas listing everything the new bicycle repair and retail business can use: tools and benches for EUR 18,000, a small delivery van for EUR 14,000, bikes held for resale for EUR 9,200, and EUR 4,800 in the business bank account. A bank has lent them EUR 22,000, and the rest has come from money and equipment they put in themselves, so that list of resources - and the story of who financed it - is already the seed of a balance sheet.

Assets Resources the business owns and uses. Office tools that stay in the workshop and bikes waiting to be sold are both assets - but they are different kinds of assets, because one is kept for use and the other is held to be sold.

Liabilities Debts and obligations owed to others (banks, suppliers, and similar creditors) that must be repaid at a stated time or over a period.

Owner's equity (equity) The portion of assets not financed by debt - what the owners have funded themselves. $\text{Equity} = \text{assets} - \text{liabilities}$. It is also a signal of the firm's wealth cushion.

Balance sheet A statement at a point in time that lists assets on one side and shows how those assets were financed through liabilities and equity on the other. The two sides must equal.

The identity must hold because every euro of asset was funded either by owners (equity) or by creditors (liabilities), and those funds are bound in whatever form the business currently holds - including cash. If assets rise, either liabilities or equity must rise by the same amount, unless another asset falls in an asset swap that leaves the total unchanged.

Balance-sheet identity

Assets = liabilities + owner's equity

Assets = resources owned and used by the business; liabilities = amounts owed to outsiders; owner's equity = owners' financing claim (assets - liabilities).

Assets		Equity and liabilities	
Office equipment	25,000	Owner's equity	24,000
Van	8,000	Bank loan	25,000
Inventory	12,500		
Cash and bank	3,500		
Total assets	49,000	Total equity and liabilities	49,000

assets = equity + liabilities (both sides always balance)

Figure 13. Balance sheet identity: every asset is financed by liabilities or by equity - the two financing sides add up to total assets.

Building Northline's opening balance sheet means summing assets first: tools and benches 18,000 + van 14,000 + inventory (bikes) 9,200 + cash 4,800 = EUR 46,000. Liabilities are the bank loan of EUR 22,000, so equity = assets - liabilities = 46,000 - 22,000 = EUR 24,000, and the check 46,000 = 22,000 + 24,000 confirms that opening totals balance at EUR 46,000 on each side.

Assets	EUR	Liabilities and equity	EUR
Tools and workshop benches	18,000	Owner's equity	24,000
Delivery van	14,000	Bank loan	22,000
Inventory (bikes for resale)	9,200		
Cash and bank	4,800		
Total assets	46,000	Total liabilities and equity	46,000

Table 9. Northline Bike Workshop - opening balance sheet (EUR)

Before you read on, pause on a small financing choice: if Northline buys EUR 1,500 of spare parts for cash, does total assets change, and what if the same purchase is on supplier credit? Paying cash for an asset is an asset swap in which one asset rises and cash falls by the same amount, so total assets stay unchanged; buying on credit raises the asset and a liability (trade payable) together, so the balance-sheet total rises. Starting from assets 46,000, liabilities 22,000 and equity 24,000, a cash purchase leaves inventory +1,500 and cash -1,500 with totals still $46,000 = 22,000 + 24,000$, while a credit purchase leaves inventory +1,500 and trade payables +1,500, so assets become 47,500, liabilities 23,500 and equity 24,000 - the same economic item, but different financing changes the totals.

Assets are also split by how long they stay in the business. Fixed or non-current assets normally have a useful life of more than one year, are intended for longer-term use, and are usually harder to turn into cash quickly; examples include property, plant, buildings, machinery, office equipment kept for use, longer-term financial assets, and intangible assets such as patents, trademarks and copyrights. Current assets have higher liquidity and are normally used up, sold or converted within a year - inventory (merchandise not yet sold), trade receivables or debtors (money customers owe), and cash. Intangible assets cannot be touched, but they still have value for the business under the same logic as tangible ones: a workshop may own none yet, while a manufacturing group may report large patents, trademarks and licences.

Liabilities follow a similar duration split. Current liabilities are due within one year and include trade payables (trade credit to suppliers), bank overdrafts and other short-term obligations, whereas non-current liabilities last more than one year and include long-term bank loans and bonds payable. A useful structure measure is the equity ratio, which asks what share of total capital the owners themselves have financed.

Equity ratio

Equity ratio = equity / total capital

Equity = owners' claim; total capital = total assets (same as total equity + liabilities). A higher ratio means a larger share of assets was financed with the firm's own resources.

Equity usually does not have to be repaid like a loan, so a higher equity share supports independence from creditors and gives a larger cushion against over-indebtedness if the firm makes a loss. One classification trap worth keeping in mind is that office computers used in the business are not the same line as computers held for resale: the physical object type can be identical, yet one is a fixed asset and the other is inventory (a current asset). In exams, section 6.1 often signals a point-in-time snapshot, the identity assets = liabilities + equity, a non-current versus current classification, or a cash purchase that leaves the balance-sheet total unchanged - and valuation is constrained by rules, so firms cannot freely invent asset values.

6.2 Income statement, cash flow and depreciation

One statement is never enough on its own. Northline's end-of-year balance sheet may show stronger equity and more cash than on day one, yet that still does not say how much was sold, what it cost to serve customers, or which cash movements came from repairs versus a new loan, so performance over a period needs the full financial statement set.

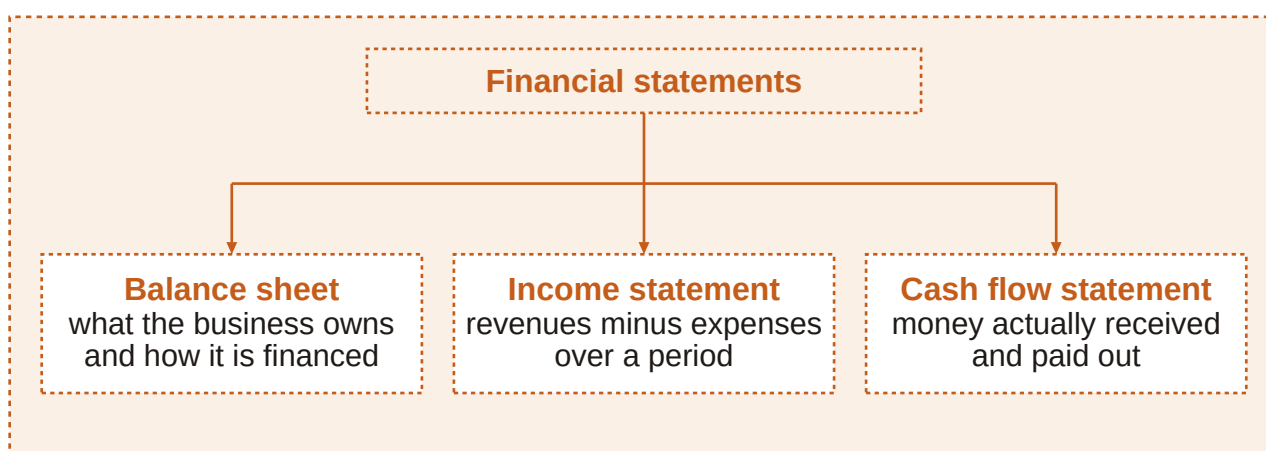


Figure 14. Three components of the financial statement: balance sheet (stock at a date), income statement (revenues, costs and expenses over a period), cash flow statement (cash in and cash out over a period).

Financial statement The package that reveals performance over a period: balance sheet + income statement (profit and loss account) + cash flow statement. The balance sheet alone cannot show turnover or production costs.

Income statement (profit and loss account) A period summary of revenues, costs and expenses. If revenues exceed costs and expenses -> profit. If costs and expenses exceed revenue -> loss.

Revenue is income from selling goods or services (cash or receivables), costs are resources consumed to produce those goods or services, and other expenses such as rent, admin

wages, energy and depreciation also reduce the result - so profit is not "cash left in the till" but the accounting surplus for the period. In a simple year for Northline, sales revenue of EUR 268,000 against costs and expenses of EUR 214,000 (materials, wages, rent, energy, depreciation and other) gives profit = $268,000 - 214,000 = \text{EUR } 54,000$, and that EUR 54,000 profit increases equity if it is retained in the business. A loss would lower equity instead, and you can spot the same economic story either in the income statement or in the change in equity between two balance-sheet dates.

Depreciation Recognition that fixed assets lose value as they are used up over time. Without it, balance-sheet asset values would be overstated and profit would ignore the wearing-out of equipment.

Straight-line logic spreads the depreciable cost evenly across the expected useful life so each year carries the same charge; land is not depreciated, and depreciation is an expense in the income statement that - unlike wages or energy - does not cause a cash payment in the period when it is charged.

Straight-line depreciation (annual)

Annual depreciation = $(\text{cost} - \text{residual value}) / \text{expected useful life}$

Cost = purchase price of the fixed asset; residual value = expected value at the end of useful life (often zero in simple cases); expected useful life = years the asset will be used. If residual value is zero: annual depreciation = cost / useful life.

For a diagnostic laptop costing EUR 2,400 with a three-year useful life and residual EUR 0, annual depreciation = $2,400 / 3 = \text{EUR } 800$, so book values fall to EUR 1,600 after year 1, EUR 800 after year 2 and EUR 0 after year 3. For the van costing EUR 14,000 with a five-year useful life and residual EUR 2,000, depreciable cost is 12,000 and annual depreciation = $12,000 / 5 = \text{EUR } 2,400$. Each year the income statement includes these charges even though the cash outlay happened when the assets were bought, and carrying (book) value equals cost minus accumulated depreciation. The cash-flow trap is never to treat depreciation as a cash outflow in the year it is charged - the cash left when the asset was purchased as an investing outflow then.

Expenditures vs expenses Expenditures are payments to purchase assets (non-current or current). Expenses are costs that have expired or been used up to produce the goods or services sold - for example COGS, salaries, marketing, interest, insurance and rent.

An expenditure is not yet an expense when Northline spends EUR 12,000 cash on a new alignment machine (an investing cash outflow): if useful life is six years and residual is EUR 0, annual depreciation expense = $12,000 / 6 = \text{EUR } 2,000$, so in year 1 cash fell by EUR 12,000 while the income statement expense is only EUR 2,000. Buying the machine is the expenditure; using it up over time creates the expenses.

Assets	EUR	Liabilities and equity	EUR
Tools and equipment (after depreciation; plus additions)	31,000	Owner's equity	78,000
Delivery van (carrying value)	11,600	Bank loan	15,000
Inventory	11,400	Trade payables	8,500
Trade receivables	6,800		
Cash and bank	40,700		
Total assets	101,500	Total liabilities and equity	101,500

Table 10. Northline - simplified year-end balance sheet after a profitable first year (EUR)

Cash flow statement Shows cash flowing into and out of the business during the period. Positive cash flow is not the same as profit.

Cash movements are grouped by purpose. Operating activities cover core business cash - customers paying, and payments to suppliers and wages linked to operations - and ideally this section is positive, because it is the most important signal of day-to-day health. Investing activities show cash spent on long-term assets or received from selling them, and a negative figure often means the firm invested rather than that it failed. Financing activities cover cash from investors or creditors and outflows for interest, dividends or debt repayment.

Cash change and profit can diverge sharply. Opening cash of EUR 4,800 and closing cash of EUR 40,700 imply a cash increase of EUR 35,900 in the same year that profit was EUR 54,000, and the gap exists because profit includes non-cash depreciation and accruals (credit sales, unpaid bills) while cash only counts actual movements - and because investing and financing cash also move the bank balance. A firm can therefore be profitable and still short of cash, or raise cash by borrowing without matching profit. Classifying a few movements makes the split concrete: customers settle EUR 38,000 of receivables (operating inflow), parts paid in cash EUR 16,000 (operating outflow), a new bench system bought for EUR 9,000 cash (investing outflow), loan principal repaid EUR 7,000 (financing outflow), and owners' drawings or dividend EUR 5,000 (financing outflow), so the net from these items is $(38,000 - 16,000) - 9,000 - 7,000 - 5,000 = +\text{EUR } 1,000$. Exam questions for 6.2 typically ask you to separate period from point-in-time, remember that profit is not cash, classify operating / investing / financing flows, compute straight-line depreciation and carrying value, spot that depreciation is an expense without a same-period cash payment, and keep expenditure (buy the asset) distinct from expense (use the asset up).

6.3 Reading balance sheets and income statements

Reading statements needs caution and then purpose, because asset values depend on valuation rules and depreciation can push book values below what an asset might fetch in a private sale - yet comparing structure over time and against similar firms still answers sharp questions about how the business is built and how it performed. On the balance sheet you ask what the asset mix looks like (mostly non-current or current, and whether that is typical), how financing has developed (equity path, more long-term or short-term liabilities), and whether non-current assets are covered by long-term finance (equity plus non-current liabilities). On the income statement you ask whether sales rose and whether cost of sales moved in line, and how gross profit and operating profit developed.

Cedar Circuit Components provides a sample statement of financial position in EUR thousands. Property, plant and equipment moves from 388 in Year 1 to 412 in Year 2; intangible assets from 52 to 48; other non-current assets from 30 to 35; so non-current assets rise from 470 to 495. Inventories move from 78 to 92, trade and other receivables from 64 to 71, and cash and cash equivalents from 86 to 118, so current assets rise from 228 to 281 and total assets from 698 to 776. On the financing side, share capital stays at 120 while reserves and retained earnings rise from 214 to 268, taking total equity from 334 to 388; non-current financial liabilities fall from 230 to 210 and other non-current liabilities rise from 24 to 28, so non-current liabilities fall from 254 to 238; trade and other payables rise from 78 to 96 and other current liabilities from 32 to 54, so current liabilities rise from 110 to 150; total liabilities therefore move from 364 to 388, and total equity and liabilities match total assets at 776 in Year 2 (698 in Year 1).

Reading that structure, $\text{non-current assets} / \text{total assets} = 495 / 776 \approx 63.8\%$, a plant-heavy mix typical of manufacturing, while the equity ratio $= 388 / 776 = 50\%$ means half of assets are financed by equity. Long-term finance $= \text{equity } 388 + \text{non-current liabilities } 238 = 626$, which exceeds non-current assets of 495, so long-term assets are covered by long-term resources, and cash of 118 exceeds trade payables of 96, so bills could be cleared from cash alone if needed. Overall the structure looks manufacturing-typical, moderately equity-funded, and matched on the long-term side.

Cedar's sample income statement (also EUR thousands) shows revenue rising from 520 to 640, cost of sales from 430 to 498, and gross profit from 90 to 142. Distribution costs move from 24 to 28, general and administrative costs from 30 to 36, and other operating result from 2 to 4, so operating result (EBIT) rises from 38 to 82. Finance costs net fall from 16 to 14, profit before tax rises from 22 to 68, income taxes from 8 to 16, and profit for the year from 14 to 52.

Cost of sales (COS) / cost of goods sold (COGS) Costs tied directly to producing the products sold - direct materials, direct production labour and manufacturing overhead. Not included: administration, shipping to customers, or sales-personnel costs.

Gross profit Earnings after deducting direct production costs from revenue - before operating expenses such as distribution and administration. It focuses on production-related margin.

EBITDA and EBIT EBITDA (earnings before interest, taxes, depreciation and amortisation) looks at operating performance with a different treatment of non-cash charges. Deducting depreciation and amortisation yields EBIT (earnings before interest and taxes) - the operating result line used widely in analysis.

Reading down the income statement, revenue minus COS gives gross profit, then operating expenses and other operating items shape the path toward EBIT, while finance income and costs sit below operating result and taxes sit near the bottom - each layer answering a different question about production margin, operating strength, then financing and tax effects. For Cedar, revenue rose by $640 - 520 = 120$ (about +23.1%) while cost of sales rose by $498 - 430 = 68$ (about +15.8%), so revenue grew faster than COS and gross profit jumped from 90 to 142, while EBIT more than doubled from 38 to 82. Top-line growth with a favourable COS relationship improved both gross profit and operating result; if instead revenue had risen only 10% while COS rose 18%, you would expect gross profit pressure and a story about weaker production efficiency or higher input prices. A common misreading is to put distribution or admin staff costs into COGS, which makes gross profit look worse for the wrong reason and confuses production performance with broader operating costs. Exam cues for 6.3 include calculating structure percentages, testing long-term matching, identifying what belongs in COS, interpreting gross profit versus EBIT, and remembering that statements need careful valuation awareness - and the durable lesson is that the balance sheet shows mix of assets and financing while the income statement shows revenue, costs and layered profits, with COS production-tied, gross profit as pre-operating-expense margin, and EBIT as the operating result, always stronger when compared across periods or peers than as a single lonely number.

6.4 Who uses accounts - financial vs management accounting

Two rooms ask two different questions of the same business. In the workshop office Mira wonders whether specialty helmets cover their true cost if priced at EUR 49, while at the bank the loan officer asks whether Northline can service debt and whether last year's accounts were reliable - same firm, different information jobs.

Users of accounts Internal users include owners, managers and employees - they care whether the business is thriving, whether costs need cutting, and whether the investment earns a return for the risk taken. External users include tax authorities, suppliers, competitors, investors, banks and the media.

Those users are served by two types of accounting. Management (managerial) accounting is built for managers inside the firm and supports decisions about where to cut costs, how to set prices and how to allocate resources, often with more detailed and forward-looking figures for internal use. Financial accounting produces statements such as the balance sheet and income statement for decision makers inside and outside the firm - owners, banks, tax authorities, investors - appears in annual reports, and for large firms often means selected figures published on websites as well.

Auditing For some companies it is mandatory that an independent firm of accountants checks the accounts for authenticity. The outcome appears in the annual report.

Managers need both: management accounting for operational choices, and financial accounting because banks, tax authorities and investors will judge the firm on those statements. On the same decision day, a management pack might show estimated cost per custom bike build of EUR 1,120, a proposed list price of EUR 1,490, and expected monthly volume if price rises by EUR 40, while the financial pack for the bank shows equity EUR

78,000, remaining loan EUR 15,000, last year's profit EUR 54,000 and audited year-end statements - and tax authorities use the financial accounts to assess taxable profit. Price design therefore leans on management accounting, whereas external financing and tax lean on financial accounting; management accounting does not replace the balance sheet and income statement, because external parties and many mandatory duties rely on financial accounting. Exam recognition for 6.4 is mainly matching a user to internal versus external, deciding whether a task is management accounting (pricing, cost cutting) or financial accounting (published statements, bank review), and knowing that auditing is independent verification for authenticity.

6.5 Analysis of financial statements

Once Cedar's statements are complete, a lender still wants sharper answers: can the firm pay bills on time, does profit justify the capital tied up, does stock sit too long, and how heavily is it geared with debt? Ratios turn raw lines into comparable answers and cluster into liquidity (solvency in the short term), profitability (profit relative to capital or size), financial efficiency (how well resources generate activity), and financial structure or gearing (equity versus debt). Because ratios vary sharply by industry, you compare peers in the same sector or one firm over several years, and you always know which definition was used before you compare.

The ratio map for this course pairs each theme with a measure and a core idea: liquidity uses working capital (current assets left after current liabilities), the current ratio (current assets per EUR 1 of current liabilities) and the acid-test ratio (the same idea excluding inventory); profitability uses ROE (EBIT relative to equity) and ROCE (EBIT relative to capital employed); efficiency uses asset turnover (turnover per EUR 1 of average assets) and inventory turnover (how many times stock is sold or replaced); and gearing or structure uses the equity ratio (equity as a percentage of total assets) and the debt ratio (total debt as a percentage of total assets).

Liquidity starts with an intuition: imagine every short-term claim lands at once, and ask whether the firm's relatively liquid resources - cash, receivables, inventory - can cover those claims and leave anything over. That leftover idea is working capital; retailers with heavy stock often need more of it than a service firm with little inventory, and high inventory can serve customers quickly while also tying up money, needing space and risking obsolescence.

Working capital

Working capital = current assets - current liabilities

Current assets = short-term resources (cash, receivables, inventory, etc.); current liabilities = obligations due within one year. Positive working capital means current assets exceed current liabilities.

Current ratio (working capital ratio)

Current ratio = current assets / current liabilities

Same inputs as working capital, expressed as a ratio. Should exceed 1; a common comfort band discussed in the course is about 1.5 to 2.

Acid-test ratio

Acid-test ratio = (current assets - inventory) / current liabilities

Removes inventory because stock may not convert to cash when needed. A stricter liquidity test; still preferably above 1.

For Cedar in Year 2 (EUR thousands), current assets = 281, inventory = 92 and current liabilities = 150, so working capital = $281 - 150 = 131$ (positive), the current ratio = $281 / 150 \approx 1.87$ (above 1 and inside the 1.5-2 comfort discussion), and the acid-test = $(281 - 92) / 150 = 189 / 150 = 1.26$ (still above 1). Short-term cover looks adequate on these measures, though you still judge against industry peers. A short-term loan, overdraft or extra trade credit can ease a cash squeeze, but it raises current liabilities and can shrink working capital, whereas improving both cash and working capital more sustainably usually means adding equity or longer-term borrowing. A current ratio above 1 is not automatically "safe" if most current assets are slow-moving inventory - that is why the acid test exists - and fixing cash with short-term borrowing can worsen the working-capital picture. Liquidity exam work usually asks you to compute working capital, current ratio and acid test, interpret "greater than 1", explain why inventory is dropped in the acid test, and link short-term borrowing to lower working capital.

Profitability starts from a similar intuition: a profit of EUR 50,000 sounds fine until you ask relative to how much capital was risked, so profitability compares profit to a size base - equity, capital employed, or turnover - and tiny profits on huge capital look weak.

Return on equity (ROE)

ROE = profit before tax and interest (EBIT) / equity

EBIT = operating profit before interest and tax (as used in this course's ROE definition); equity = owners' financing on the balance sheet.

Return on capital employed (ROCE)

ROCE = profit before tax and interest (EBIT) / capital employed

Capital employed $\approx$ equity + non-current liabilities, or equivalently assets - current liabilities. Average capital employed (beginning and end of year, $\div 2$) may be used instead of the year-end figure.

For Cedar Year 2 (EUR thousands), EBIT = 82 and equity = 388, so ROE = $82 / 388 \approx 21.1\%$. Capital employed = assets - current liabilities = $776 - 150 = 626$ (also equity 388 + non-current liabilities 238), so ROCE = $82 / 626 \approx 13.1\%$; if average capital employed were 640, ROCE =

$82 / 640 = 12.8\%$. Those returns look meaningful only beside peer ROCE/ROE and Cedar's own trend.

Efficiency asks how well resources generate activity: asset turnover asks how many euros of sales each euro of assets supports, while inventory turnover asks how often stock is sold or used up and replaced, with high turnover meaning money is not trapped in the warehouse for long.

Asset turnover

Asset turnover = turnover / average assets

Turnover = sales revenue for the period; average assets = (assets at beginning + assets at end) / 2. An alternative form uses average net assets (assets - long-term / non-current liabilities).

Inventory (stock) turnover

Inventory turnover = cost of sales / average inventory

Cost of sales = COS/COGS for the period; average inventory = typical stock level over the year. Result is 'times per year'. Days in inventory $\approx 365 / \text{inventory turnover}$.

For Cedar (EUR thousands), turnover (revenue) = 640 with assets of 698 in Year 1 and 776 in Year 2, so average assets = $(698 + 776) / 2 = 737$ and asset turnover = $640 / 737 \approx 0.87$ - about EUR 0.87 of sales per EUR 1 of average assets. Using net assets instead, Year 2 net assets = $776 - 238 = 538$ and Year 1 = $698 - 254 = 444$, so average = 491 and turnover / average net assets = $640 / 491 \approx 1.30$. Average inventory = $(92 + 78) / 2 = 85$, so inventory turnover = $498 / 85 \approx 5.9$ times per year, or roughly $365 / 5.9 \approx 62$ days. Stock renews about six times a year, and asset productivity must still be judged against manufacturing peers.

Gearing, or financial structure, asks how heavily the asset base leans on debt versus owners' funds: high debt can amplify returns but also raises dependence on creditors and loss pressure, and the course measures that structure with the equity ratio and the debt ratio.

Equity ratio

Equity ratio = equity / total assets

Equity = total equity; total assets = balance-sheet total (equals equity + liabilities).

Debt ratio

Debt ratio = total debt / total assets

Total debt = total liabilities (amounts owed); total assets = balance-sheet total. Equity ratio and debt ratio together describe the financing split (they add to 1 if debt means all liabilities).

For Cedar Year 2, equity = 388, total assets = 776 and total liabilities (debt) = 388, so the equity ratio = $388 / 776 = 50\%$ and the debt ratio = $388 / 776 = 50\%$ - half equity, half debt financing, a balanced structure on these definitions that still needs sector comparison. If

Cedar financed a new plant mostly with short-term overdraft instead of equity or a long-term loan, liquidity would worsen first as current liabilities rose and working capital fell, while gearing would also shift toward more debt, so both clusters would move, not only one. Rounding the Year 2 picture: working capital EUR 131k (positive short-term buffer), current ratio 1.87 (above 1, near the 1.5-2 band), acid-test 1.26 (above 1 without inventory), ROE 21.1% (return on owners' funds), ROCE 13.1% (return on long-term capital), asset turnover 0.87 (sales per EUR 1 of assets), inventory turnover 5.9x (~62 days in stock), equity ratio 50% and debt ratio 50%. Liquidity, profitability, efficiency and gearing answer different questions about the same statements, so a firm can look profitable on ROE yet strained on the acid test, or highly efficient at turning assets while heavily geared - which is why you read the set together, always comparing within industry or over time and knowing the definition used.

Chapter recap

- The balance-sheet identity is $\text{assets} = \text{liabilities} + \text{equity}$; classify fixed/non-current versus current assets and liabilities, and treat equity as the owners' financing share.
- Financial statements combine a balance sheet (point in time), an income statement (period profit or loss) and a cash flow statement (operating, investing and financing cash).
- Straight-line depreciation spreads depreciable cost over useful life as an expense without a same-period cash payment; expenditures buy assets, while expenses are used-up costs.
- Reading skill means structure and matching on the balance sheet, and COS/COGS, gross profit, EBITDA and EBIT layers on the income statement.
- Management accounting supports internal decisions; financial accounting informs internal and external users; auditing checks authenticity where required.
- Analysis moves from intuition to formula to interpretation across liquidity (working capital, current ratio, acid test), profitability (ROE, ROCE), efficiency (asset turnover, inventory turnover) and gearing (equity ratio, debt ratio).

End of the Full Course theory

You have walked the BBE Path from scene to exam language across Chapters 2–6. Continue with practice tasks for the chapter you have just studied.